Observation Theory: Geometry, Distortion, and Reliability as Properties of Observation

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Abstract

We develop Observation Theory, the thesis that operational geometry, distortion, and reliability are properties not of an object alone but of the relation among the object, an observer, a channel, and a resolution budget. An observer is a consumer, a map from a representation to a task output, with a metric on that output. A second-order expansion of the downstream distortion isolates one controlling quantity: the reconstruction error read through the consumer, the trace of the error second moment weighted by a positive semidefinite read operator, the pullback of the output metric along the consumer. Ordinary reconstruction is the isotropic corner where this operator is the identity, the home of Shannon's Gaussian mean-squared-error rate distortion and Gauss least squares. We prove that the read operator is the pullback geometry of distinguishability, inducing a quotient by what the consumer cannot separate; that its read subspace is recoverable from black-box consumer queries under a specified observation distribution; and that it defines a consumer relative rate distortion function whose achievability and converse we establish for a general source, with a coding theorem for the true nonlinear consumer divergence that reduces exactly to output coding. We instantiate it for attention, gradient optimization, and retrieval, and position it as the identifiable geometric middle term between Shannon rate distortion and the information bottleneck. We determine, for a *Gaussian source*, the complete two-stage rate region for two read operators—successively refinable exactly when their optimal error covariances nest—extend it to k observers, and price a misidentified observer: mis-weighting a read direction costs a bounded rate tax, omitting one erects an irreducible distortion floor. Every empirical claim is sealed-registered against a preset bar; of the five core predictions (GO-1–GO-5) four are confirmed, one refuted, and a corrected trained-model test passed.

Index Terms

Distinguishability, finite blocklength, indirect source coding, information bottleneck, information geometry, neural network compression, quantization, rate–distortion theory, reproducibility, source–channel separation, source coding, successive refinement.

I. Introduction

Shannon fixed a distortion measure and asked for the fewest bits meeting it [1], [2]; Gauss fixed the same measure and asked for the least-squares estimate [3]. This paper reads their common assumption as the variable of interest: the error that matters is not the one a code makes but the one a consumer *reads*, and different consumers read different subspaces of the same error. Once the distortion is that of the task, Euclidean reconstruction is one corner—the isotropic observer $P_{\mathcal{C}} = I$—and any anisotropic or rank-deficient $P_{\mathcal{C}}$ reads the error differently.

The idea that one should code a source to serve a *functional* of it, not the source, is not new: it is the indirect (remote) rate–distortion problem [5], [6], [7]; its lossless, discrete form is coding for computing and graph entropy [8], [9]; its modern face is the rate–distortion–perception tradeoff [10] and task-oriented / semantic communication [11]; and its deployed form, for neural networks, is loss-aware (Hessian-weighted) compression, from Optimal Brain Damage/Surgeon [16], [17] to Hessian-aware quantization and GPTQ [18], [19]. What is missing across these is a single, *identifiable geometric* object that names the task's metric and lets one predict, recover, and test it. Observation Theory supplies it: the read operator $P_{\mathcal{C}}$, the consumer's local output metric, sitting as the geometric middle term between Shannon's Gaussian Euclidean-MSE specialization ($P_{\mathcal{C}} = I$) and the information bottleneck [12] (an arbitrary, non-geometric relevance variable), as Table I summarizes. We contribute (i) the map placing these literatures on one axis; (ii) $P_{\mathcal{C}}$'s *read-subspace recoverability*—recovery of the read subspace from black-box consumer queries under the specified probing model, validated blind against a planted ground truth, which the quantization literature (estimating the Hessian from calibration data) never validates against truth; (iii) an ex-ante extreme-value reliability certificate; (iv) budget-relativity of the verdict; and (v) a registration-first empirical program—sealed predictions, ex-ante bars, a refuted prediction reported as prominently as the confirmations—whose recurring demonstration is a single code whose downstream verdict *inverts* when nothing changes but the consumer; (vi) the complete two-stage rate region for a Gaussian source under two read operators—with successive refinability (the optimal error covariances nest) and a closed-form max-det rate loss as its corner, extended to k observers (§X)—together with a finite-blocklength reduction (with, in the Gaussian case, a dispersion counting read dimensions) and an observer source–channel separation (§IX); and (vii) for a Gaussian source, an operational price for a *mis*identified read operator: a bounded rate tax for mis-weighting, an irreducible distortion floor for omission (Appendix C). The three companion works are Paper I (the angular observer and the spectral budget-inversion

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Companion software, the sealed pre-registrations, and the claim ledger that govern every empirical result are publicly available at https://github.com/ahb-sjsu/geometric-observation. A continuous-integration workflow re-runs the reproducible experiments and verifies the theoretical constructions on each commit.

TABLE I

OBSERVATION THEORY AS THE IDENTIFIABLE GEOMETRIC MIDDLE TERM BETWEEN TWO CLASSICAL EXTREMES. THE READ OPERATOR $P_{\mathcal{C}}$ IS THE AXIS.

Framework	$P_{\mathcal{C}}$	**Character**
Shannon rate–distortion (Gaussian MSE)	I (isotropic)	fixed, task-agnostic; the reconstruction corner where Gauss least squares lives
Observation Theory	$J^\top GJ$	geometric, positive semidefinite, water-fillable, with its read subspace *recoverable* from consumer queries under a specified probe
Information bottleneck	—	an arbitrary, non-geometric relevance variable

mechanism) [20], Paper II (geometric key quantization and the reconstruction-is-not-the-target negative) [21], and the claim ledger [22]; this is their information-theoretic synthesis.

II. THE CONTROLLING QUANTITY

Definition 1 (Instance)**.** *An instance is $(X, \mathcal{C}, \mathcal{Q})$: a representation with vectors $x \in \mathbb{R}^d$, a* consumer *$\mathcal{C} : \mathbb{R}^d \to \mathbb{R}^m$ with a divergence D_Y on its outputs, and a family $\mathcal{Q}$ of lossy codes producing $\hat{x} = x + \delta$. The downstream distortion is the pointwise random quantity $d_{\mathcal{O}} := D_Y(\mathcal{C}(x), \mathcal{C}(\hat{x}))$, measured on the consumer output, never assumed from the ambient metric of X; write $\bar{d_{\mathcal{O}}} := \mathbb{E}[d_{\mathcal{O}}]$ for its expectation over the instance.*

> **Example 1** (The attention head as observer)**.** A concrete instance to keep in mind is a single attention head. Here X is the set of key vectors, the consumer $\mathcal{C}$ is the softmax-attention map $x \mapsto \mathrm{softmax}(q^\top x/\sqrt{d})$ that the head's queries q induce, the output divergence D_Y is the Kullback–Leibler divergence between attention distributions, and $\mathcal{Q}$ is a family of low-bit key quantizers. The read operator's feature-space factor will turn out to be the query second moment $P_{\mathcal{C}} = \mathbb{E}[qq^\top]$ (§III)—the reduced form, under decorrelated or uniform-weight queries, of the exact coupled softmax–Fisher operator of Prop. 2—and our experiments make the consequences concrete: two 4-bit key quantizers with *identical* reconstruction error rank in opposite order under the head's softmax-KL, and the ranking *flips* with the query bank (12/12 seeds, §IV); a blind probe recovers this $P_{\mathcal{C}}$'s top subspace and names the winner before any bits are spent (§V); and on a real Llama-3.2-3B layer the same blind prediction is right on all 16 instances while reconstruction, held equal, cannot discriminate (2/16; §XIII). One head, carried from definition to trained model.

Proposition 1 (Consumer-projected distortion)**.** *Let D_Y be twice differentiable with $D_Y(u, u) = 0$, $\nabla_v D_Y(u, v)|_{v=u} = 0$, and* local output metric *$G(u) = \nabla_v^2 D_Y(u, v)|_{v=u} \succeq 0$ (the Fisher information metric when D_Y is the KL divergence or a kindred statistical divergence, a generic local metric otherwise); let $J(x) = \partial\mathcal{C}/\partial x$. For error $\delta = \hat{x} - x$ with conditional* second moment *$\Sigma_\delta(x) = \mathbb{E}[\delta\delta^\top \mid x]$, the leading-order expected distortion is*

$$\bar{d_{\mathcal{O}}} \;=\; \tfrac{1}{2}\,\mathbb{E}_x \operatorname{tr}\big(P_{\mathcal{C}}(x)\,\Sigma_\delta(x)\big) \;+\; r, \qquad P_{\mathcal{C}}(x) = J(x)^\top G(\mathcal{C}(x))\,J(x) \succeq 0,$$

so $P_{\mathcal{C}}$ is the pullback of the local output metric G along $\mathcal{C}$. Making the asymptotics explicit: for a small-noise family $\delta_\varepsilon = \varepsilon Z$ with $\mathbb{E}\|Z\|^2 < \infty$ and the second-order Taylor remainder dominated uniformly on a neighborhood of the diagonal, $\bar{d_{\mathcal{O}}} = \frac{\varepsilon^2}{2}\,\mathbb{E}[Z^\top P_{\mathcal{C}}(X)Z] + o(\varepsilon^2)$ (i.e. $r = o(\varepsilon^2)$). When $\Sigma_\delta(x)$ is uncorrelated with the point-dependent weight $P_{\mathcal{C}}(x)$ this factors as $\frac{1}{2}\operatorname{tr}(P_{\mathcal{C}}\Sigma_\delta)$ with $P_{\mathcal{C}} = \mathbb{E}_x[P_{\mathcal{C}}(x)]$, $\Sigma_\delta = \mathbb{E}_x[\Sigma_\delta(x)]$. The controlling object is the second *moment, not the covariance: pointwise $\operatorname{tr}(P_{\mathcal{C}}(x)\Sigma_\delta(x)) = \operatorname{tr}(P_{\mathcal{C}}(x)\operatorname{Cov}[\delta \mid x]) + \mu_\delta(x)^\top P_{\mathcal{C}}(x)\mu_\delta(x)$ with $\mu_\delta(x) = \mathbb{E}[\delta \mid x]$, then averaged over x, so a biased code's offset is* part *of the distortion, not averaged away—which is why quantizer bias is first-class (§III). Reconstruction distortion $\mathbb{E}\|\delta\|^2 = \operatorname{tr}(\Sigma_\delta)$ is the* isotropic *corner $P_{\mathcal{C}} = I$.*

Proof. Expand $\mathcal{C}(x + \delta) = \mathcal{C}(x) + J\delta + O(\|\delta\|^2)$ and D_Y to second order about its minimum; the value and gradient terms vanish, leaving $\frac{1}{2}\delta^\top(J^\top GJ)\delta + o(\|\delta\|^2)$. Take $\mathbb{E}_x$; the factored form requires the stated uncorrelatedness (for real quantizers both J and Σ_δ depend on x, so the exact object is the x-averaged trace). □

We keep $\mathbb{E}_x \operatorname{tr}(P_{\mathcal{C}}(x)\Sigma_\delta(x))$ as primary and use the factored $\operatorname{tr}(P_{\mathcal{C}}\Sigma_\delta)$ as the convenience it is. (We drop the harmless $\frac{1}{2}$.)

Remark (When the factored form is exact). Because the rest of the paper leans on $\operatorname{tr}(P_{\mathcal{C}}\Sigma_\delta)$, it is worth stating when it holds. The factorization is *exact* whenever the error second moment $\Sigma_\delta(x)$ does not co-vary with the read operator $P_{\mathcal{C}}(x)$ across the support—in particular for a *linear* consumer, where $P_{\mathcal{C}}$ is constant. It holds to good approximation for a *fixed-codebook* quantizer whose per-coordinate error statistics are roughly homogeneous over x (uniform or per-channel key quantization on a representation whose read subspace is stable), which covers the codes studied in §IV. When the consumer is strongly point-dependent—a deep network whose $P_{\mathcal{C}}(x)$ swings with the activation—the coupled expectation $\mathbb{E}_x \operatorname{tr}(P_{\mathcal{C}}(x)\Sigma_\delta(x))$ must be retained; Appendix B then gives the exact, unfactored treatment.

TABLE II
THE READ OPERATOR $P_{\mathcal{C}}$ INSTANTIATED FOR THREE CONSUMERS. ORDINARY RECONSTRUCTION IS THE ISOTROPIC CORNER $P_{\mathcal{C}} = I$.

Consumer	**Output divergence** D_Y	**Read operator** $P_{\mathcal{C}}$	**Recovered from**
Softmax attention	Kullback–Leibler	query second moment $\mathbb{E}[qq^\top]$ (coupled expectation)	finite-difference Jacobian probe of the softmax output
Gradient optimizer	quadratic step error	Hessian H (Gauss–Newton curvature if nonconvex)	calibration curvature estimate
Retrieval ranking	margin / rank loss	subspace projector $\mathbb{E}[qq^\top]$	query-bank second moment
Reconstruction	squared Euclidean	identity I (isotropic corner)	— (task-agnostic)

Distinguishability, and the observer's quotient

The form $\delta^\top P_{\mathcal{C}}(x)\delta = (J\delta)^\top G\,(J\delta)$ is not, at root, a compression cost; it is a *distinguishability*. It measures how far the consumer's outputs $\mathcal{C}(x)$ and $\mathcal{C}(x+\delta)$ separate in the output metric G, and when G is the Fisher information metric of the output distribution, $\delta^\top P_{\mathcal{C}}\delta$ *is* the local statistical distinguishability (the leading KL / χ^2) of those outputs. So $P_{\mathcal{C}}$ is the pullback through $\mathcal{C}$ of the geometry of distinguishability: Observation Theory is pullback geometry.

Two consequences follow, and both recur below. First, define the global operational equivalence $x \sim_{\mathcal{C}} x' \iff D_Y(\mathcal{C}(x), \mathcal{C}(x')) = 0$ (statistical indistinguishability of the outputs). Its infinitesimal shadow is $\ker P_{\mathcal{C}}(x) = \ker\big(G^{1/2}J\big)$, which equals $\ker J$ when $G \succ 0$ on $\operatorname{im} J$ but is larger when G is only semidefinite—the output can move parametrically yet stay indistinguishable. Under a constant-rank / integrability condition on $P_{\mathcal{C}}(x)$ this null distribution integrates to leaves tangent to $\ker P_{\mathcal{C}}(x)$; that the *global* divergence vanishes along each leaf—so the leaves are exactly the fibers of $\sim_{\mathcal{C}}$—needs a further compatibility condition, namely $D_Y(\mathcal{C}(x), \mathcal{C}(x')) = 0$ along them (equivalently, $\mathcal{C}$ factors through $X/\!\sim_{\mathcal{C}}$). Where it holds, the consumer's true object is the quotient $X/\!\sim_{\mathcal{C}}$, not X: perturbations in $\ker P_{\mathcal{C}}$ exist in the representation but have *no operational consequence*. In the fixed-$P_{\mathcal{C}}$ model of §VIII this quotient is represented by ΠX and Prop. 7 gives the zero-distortion rate $R_{\mathcal{C}}(0) = H(\Pi X)$; the point-dependent case is settled unconditionally in Appendix B (Cor. 7: $R_{\mathcal{C}}(0) = H(\mathcal{C}(X))$, via the true divergence, needing none of the compatibility conditions above)—rate is expended only on distinctions the consumer can resolve. Second, $\ker P_{\mathcal{C}}$ is gauge-like *but only relative to* $\mathcal{C}$: another consumer may read those directions, so it is not a true gauge redundancy (which would be invisible to every admissible observer). One observer's noise is another's signal; one observer's state variable is another's gauge freedom. Adjoining the budget of §VII, an observer is a triple $\mathcal{O} = (\mathcal{C}, G, B)$—readout, output metric, and resolution—and the geometry it resolves is relative to all three.

> **Principle of observational relativity.** Operational fidelity, distortion, and resolved geometry are not properties of the object alone; they are properties of the relation among object, observer $(\mathcal{C}, G)$, channel, and budget, and are ill-posed until all are specified. Observation induces geometry; each observer pulls back a distinct operative geometry and quotient.

The rest of the paper derives $P_{\mathcal{C}}$ for concrete consumers, shows it recoverable from queries, and studies what the channel (§VI) and budget (§VII) do to the geometry it induces.

III. DERIVATION OF THE READ OPERATOR

We now derive $P_{\mathcal{C}}$ for three consumers, collected in Table II. The first is the attention head of Example 1.

Proposition 2 (Softmax attention)**.** *Read keys $\{k_s\}_{s=1}^S$ through a query bank: logits $l_s = q^\top k_s/\sqrt{d}$, output $p = \operatorname{softmax}(l)$, $D_Y = \mathrm{KL}$. Under key perturbation δ_s,*

$$\bar{d_{\mathcal{O}}} = \tfrac{1}{2d}\,\mathbb{E}_q \operatorname{Var}^p_s\!\big(q^\top \delta_s\big) + O(\|\delta\|^3) = \tfrac{1}{2d}\,\mathbb{E}_q \operatorname{tr}\!\big(qq^\top A_q\big) + O(\|\delta\|^3), \qquad A_q = \sum_s p_s(q)\,\tilde{\delta}_s\tilde{\delta}_s^\top,$$

where $\tilde{\delta}_s = \delta_s - \sum_{s'} p_{s'}\delta_{s'}$ is the across-token p-demeaned error and A_q its $p(q)$-weighted second moment. This factors *to $\frac{1}{2d}\operatorname{tr}(Q\,\bar{\Sigma}_\delta)$, with $Q = \mathbb{E}_q[qq^\top]$ and $\bar{\Sigma}_\delta = \mathbb{E}_q[A_q]$,* only *when A_q is uncorrelated with $qq^\top$ (the proviso of Prop. 1); in general the expectation stays coupled, since p depends on q.*

Proof. The Fisher metric of softmax is $G = \operatorname{diag}(p) - pp^\top$, so $\mathrm{KL}(p\|\operatorname{softmax}(l+\Delta l)) = \frac{1}{2}\operatorname{Var}^p_s(\Delta l_s) + O(\Delta l^3)$ with $\Delta l_s = q^\top\delta_s/\sqrt{d}$. The $-pp^\top$ term is the softmax's invariance to a common logit shift; it removes the p-mean, giving $\tilde{\delta}_s$. Since p depends on q, the outer $\bar{\Sigma}_\delta$ carries the $\mathbb{E}_q$ and the final Q-factorization inherits the uncorrelatedness proviso of Prop. 1. ☐

Two consequences are the phenomena we later measure. The feature-space factor is generated by $qq^\top$, reducing to $Q = \mathbb{E}_q[qq^\top]$ under the decorrelation / uniform-weight regime above; so an isotropic bank yields an isotropic feature operator (reconstruction-like) while a subspace bank projects—the read operator, hence the winning code, changes with the query distribution. And only the *demeaned* error survives, so the across-token mean is a nuisance the softmax cannot see. This last

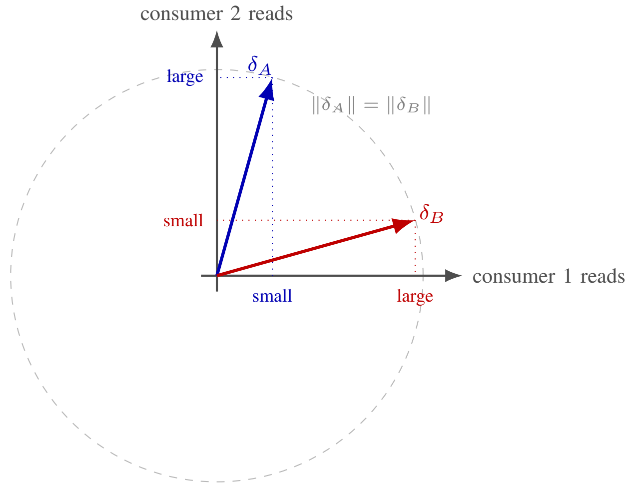

Fig. 1. The flip. Two lossy codes produce errors δ_A, δ_B of *equal* reconstruction norm (dashed circle), so reconstruction cannot separate them. Consumer 1 (horizontal read operator) sees a small δ_A projection and a large δ_B one, and prefers code A; consumer 2 (vertical read) prefers code B. The downstream verdict $\operatorname{tr}(P_{\mathcal{C}}\Sigma_\delta)$ reverses with the read operator while reconstruction remains blind—the characteristic signature of Observation Theory.

fact resolves an apparent tension with the DC-offset collapse of Paper II [21]: a truly common error component is free by shift invariance, but a per-channel DC *offset* is not a common component—it inflates the quantizer's range, wasting levels and enlarging the *visible* per-token error, which is exactly why an asymmetric zero-point (asym-NF4) repairs it at no bit cost. Invisible-to-the-consumer and free-to-encode are the same statement; range waste is a different one.

Proposition 3 (Gradient-descent optimizer; exact)**.** *Let $L(\theta) = \frac{1}{2}\theta^\top H\theta$ with $H \succeq 0$ and $g = H\theta$; the consumer maps a gradient to its updated iterate $\mathcal{C}(g) = \theta - \eta g$ at step size η, read through the loss* Bregman divergence $D_L(u,v) = L(u) - L(v) - \nabla L(v)^\top(u - v)$. *For a noisy gradient $\hat{g} = g + \delta$ the downstream distortion is,* pointwise*,*

$$d_{\mathcal{O}} = D_L(\theta - \eta\hat{g},\ \theta - \eta g) = \tfrac{1}{2}\eta^2\, \delta^\top H\delta \ \geq 0,$$

so $\mathbb{E}[d_{\mathcal{O}}] = \frac{1}{2}\eta^2 \operatorname{tr}(H\,\Sigma_\delta)$ and $P_{\mathcal{C}} = H \succeq 0$ exactly. *Measuring the update error in the loss geometry rather than the raw excess loss removes the indefinite linear cross term, so no unbiasedness qualification on δ is needed.*

This is loss-aware pruning/quantization made a metric: Hessian-weighted deletion [16], [17], [18], [19] *is* $\operatorname{tr}(H\Sigma_\delta)$, and calibration-based Hessian estimation is the production form of the probe of §V. Reconstruction is the flat case $H = I$. For a non-convex loss the Hessian may be indefinite; the metric reading then takes $P_{\mathcal{C}}$ to be the PSD Gauss–Newton / Fisher curvature in place of H.

Remark (Retrieval)*.* For score-ranking $s(q, x_i) = q^\top x_i$, the inverted-pair fraction is, to leading order for smooth rank statistics, monotone in $\operatorname{tr}(P_{\mathcal{C}}\bar{\Sigma}_\delta)/\operatorname{tr}(P_{\mathcal{C}}\bar{\Sigma}_0)$, with $P_{\mathcal{C}} = \mathbb{E}[qq^\top]$, $\bar{\Sigma}_\delta = \operatorname{Cov}_i(\delta_i)$ the across-item error covariance and $\bar{\Sigma}_0 = \operatorname{Cov}_i(x_i)$ the signal covariance (both demeaned across items); $P_{\mathcal{C}}$ is a subspace projector when queries concentrate. Rank statistics are non-smooth; we treat this instance empirically (§IV).

IV. Distortion: The Consumer Determines the Metric (GO-2)

Proposition 2 predicts that at matched reconstruction the ranking of two codes is set by $\operatorname{tr}(P_{\mathcal{C}}\Sigma_\delta)$, hence by the query distribution.

a) Dissociation.: Two 4-bit codes at an audited matched budget (spread 0.19 bit) with *identical* reconstruction (0.0938 vs 0.0934): the whole-vector code is $2.53\times$ worse on softmax-KL (12/12 seeds); a wrong-quotient control is $21\times$ worse at $\sim 5\times$ the reconstruction. Reconstruction is not the controlling quantity (*Demonstrated*).

b) Flip.: Holding reconstruction fixed and changing only the query bank, the ranking *inverts* (Fig. 1): isotropic $\Rightarrow$ code a better ($2.5\times$, 12/12); subspace $\Rightarrow$ code b better (12/12). The directly computed $\operatorname{tr}(P_{\mathcal{C}}\Sigma_\delta)$ tracks the sign on every seed in both consumers; reconstruction, a consumer-blind scalar, cannot (rank correlation 0.40 vs 1.00). This is Proposition 2 with Q swapped (*Demonstrated*).

c) Replication and out-of-sample.: The flip replicates on retrieval ranking (different representation, set-valued consumer): identical reconstruction (0.0964), ranking flips with the read subspace (log-advantage -4.70 vs $+4.65$, 12/12), $\operatorname{tr}(P_{\mathcal{C}}\Sigma_\delta)$ tracking, reconstruction blind (*Replicated*). It generalizes to optimization with $P_{\mathcal{C}} = H$ (Prop. 3): at matched reconstruction the curvature-metric distortion flips between two Hessians (12/12), tracked by $\operatorname{tr}(H\Sigma_\delta)$, reconstruction uninformative (-0.20)—and under $H = I$ reconstruction tracks *perfectly* (1.00), the derived $P_{\mathcal{C}} = I$ corner observed (*Predicted*, out-of-sample). One derivation; three read operators. These consumers are *synthetic*—the point is to validate the theory's logic (the $H = I$ positive

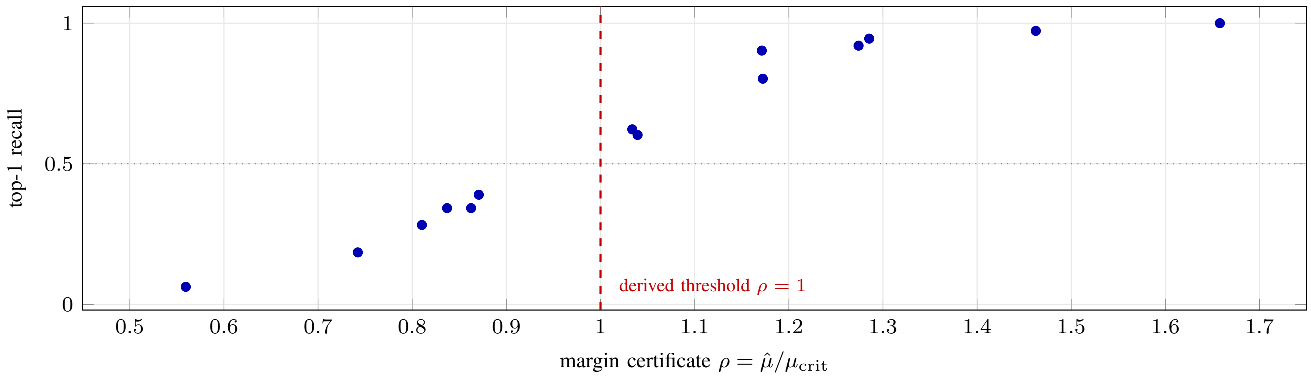

Fig. 2. Capacity (GO-3), from the committed result artifact. The derived margin certificate ρ orders top-1 retrieval recall across 14 structurally distinct corpora at Spearman correlation 0.99; the recall-$\frac{1}{2}$ crossing sits at $\rho \approx 0.95$, near the extreme-value vacuity threshold $\rho = 1$ (dashed) at which the channel is predicted to die *before* any retrieval is run.

control being the clean confirmation); the trained-model hardening (§XIII), attempted prospectively, initially missed on a non-reconstruction-matched design and is then *cleared* by a corrected, reconstruction-matched rematch on a real Llama attention layer.

V. Identifiability: the Read Subspace is Recoverable (GO-1)

Proposition 4 (Recovery of the read subspace)**.** *Let $\mathcal{C}(x) = f(Wx)$, $W \in \mathbb{R}^{r\times d}$ of rank r, f coordinatewise with $\mathbb{E}[f_i'(Wx)^2] > 0$ for every coordinate i (no saturated unit). Then $\mathbb{E}_x[J^\top J] = W^\top \mathbb{E}[\mathrm{diag}(f'^2)]W$ has rank r with column space $\mathrm{row}(W)$, so its top-r eigenvectors span the read subspace; a finite-difference estimate $\hat{J}u \approx (\mathcal{C}(x{+}hu) - \mathcal{C}(x{-}hu))/2h$ recovers it from consumer queries alone.*

We built such a probe—receiving only the callable $\mathcal{C}(\cdot)$ and the dimensions, never W, the codes, or $d_{\mathcal{O}}$ (blinding enforced in code). It recovers the planted read subspace at overlap 0.936 against a 0.059 chance baseline ($\approx 16\times$), and the *blind* $\mathrm{tr}(\hat{P}_{\mathcal{C}}\Sigma_\delta)$ predicts the flip of §IV on 12/12 seeds in both consumers with rank correlation 1.00—identical to ground truth—while reconstruction cannot (0.40) (*Predicted*). Two scope notes. Prop. 4 recovers the read *subspace* for the $f(Wx)$ model with a Euclidean output metric; recovering the *full* operator $\mathbb{E}_x[J^\top GJ]$—its eigenvalues, and a general G—is the broader program. And $P_{\mathcal{C}}$ depends on the probe-point distribution and on G, so it is identifiable from black-box consumer queries *under a specified observation distribution*, not from the callable alone. Loss-aware compression estimates such a metric from calibration data; the contribution here is validating *recovery against a planted ground truth under enforced blinding*, and composing it: from consumer queries under that distribution one predicts which code wins, before compressing anything.

VI. Capacity: a Derived Vacuity Threshold (GO-3)

Proposition 5 (Retrieval vacuity threshold)**.** *Model the $N{-}1$ distractor scores, standardized to mean 0, variance 1, as i.i.d. standard Gaussian (a reference model). Since $\Pr[M_{N-1} \le m] = \Phi(m)^{N-1}$, top-1 success $\Pr[\hat{\mu} > M_{N-1}]$ crosses $\frac{1}{2}$ at the* exact median *of the maximum,*

$$\mu_{\text{crit}} := \mathrm{med}(M_{N-1}) = \Phi^{-1}\big(2^{-1/(N-1)}\big), \qquad \textit{leading asymptotic } \sqrt{2\ln(N{-}1)},$$

where $\hat{\mu} = (\overline{\mathit{true}} - \overline{\mathit{distr}})/\mathrm{std}$ is the margin certificate and $\rho = \hat{\mu}/\mu_{\text{crit}}$; the certificate is vacuous *for $\rho < 1$.*

Derivation. $\Pr[M_{N-1} \le m] = \Phi(m)^{N-1}$; setting it to $\frac{1}{2}$ gives $m = \Phi^{-1}(2^{-1/(N-1)})$, the exact median. The Gumbel centering is $\alpha_N = \sqrt{2\ln(N{-}1)} - b_N(\ln\ln(N{-}1) + \ln 4\pi)/2$ with scale $b_N = 1/\sqrt{2\ln(N{-}1)}$; the crude $\sqrt{2\ln(N{-}1)}$ omits the $-b_N(\ln\ln(N{-}1) + \ln 4\pi)/2$ correction and so *overshoots*. Median and mean share this centering and differ at order b_N: $\mathrm{med}(M_{N-1}) = \alpha_N - b_N \ln\ln 2 + o(b_N)$ and $\mathbb{E}[M_{N-1}] = \alpha_N + \gamma b_N + o(b_N)$ (γ Euler–Mascheroni), so the mean exceeds the median by $(\gamma + \ln\ln 2)\, b_N$. The recall-$\frac{1}{2}$ crossing is the exact median $\Phi^{-1}(2^{-1/(N-1)})$—not the mean, not the leading order. $\square$

Across 14 structurally distinct corpora the certificate orders them by recall at rank correlation 0.991 (Fig. 2), beating the un-normalized margin (0.873), with a finite-width transition (*Demonstrated*). The registered harness used the *mean* $\mathbb{E}[M_{N-1}]$ (Monte-Carlo) as its μ_{crit} reference, and the recall-$\frac{1}{2}$ crossing sits at $\rho = 0.948$—slightly *below* 1 against that reference. This is expected and quantitative: the exact $\frac{1}{2}$ threshold is the *median*, below the mean by $\Theta(1/a_N)$, so the $\sim 5\%$ shortfall is largely the mean–median offset (against the exact median of this proposition the crossing sits at ≈ 1). Positive dependence among distractors *can* lower it further by reducing the effective count; heavier tails, by contrast, *raise* the maximum—the two must not be grouped. The exact $\frac{1}{2}$ crossing further assumes a *fixed* standardized true margin and i.i.d. Gaussian distractors; when the true score is itself random or correlated with the distractors, ρ is a reference certificate rather than an exact recall threshold. The exact threshold above is Gaussian. More generally, maxima in the Gumbel domain retain a Gumbel limit but have *tail-dependent* centering and scaling: for $\overline{F}(x) \asymp \exp(-x^\alpha)$ (a survival tail) the maximum grows like $(\ln N)^{1/\alpha}$, so Gaussian (α=2) and exponential (α=1) distractors do *not* share the same $\sqrt{\ln N}$ law—non-Gaussianity changes more than constants. Under suitable weak-dependence conditions an extremal index $\theta \in (0, 1]$ modifies the effective exceedance count from N to approximately θN, shifting the Gaussian centering by $\ln\theta/\sqrt{2\ln N}$ at leading order. We therefore use the Gaussian formula as a *reference certificate*, not a distribution-free threshold. The 14-corpus ordering above (Spearman $r_s = 0.991$ on clustered, non-Gaussian real embeddings) is the empirical counterpart: real structure shifts the absolute death point but preserves the ordering the certificate is used for. Capacity says, from a computable quantity, *where* the channel dies before any retrieval is run.

Using the certificate. For a practitioner, ρ is a cheap pre-flight check: estimate the standardized true-versus-distractor margin on a small calibration set, divide by the derived μ_{crit}, and read off ρ. A value below 1 warns of *predicted* vacuity under the reference model—single-stage retrieval is likely at or past its vacuity point, so budget for a re-ranking stage or a larger index before deployment—while ρ comfortably above 1 indicates headroom. Because ρ preserves the recall *ordering*, in the tested corpora it also ranked candidate corpora and index configurations *without* a full retrieval evaluation.

VII. Rate: The Operative Verdict Depends on the Budget (GO-4)

The read operator, capacity, and distortion are stated at a fixed observer budget; the budget is a rate, and it inverts the verdict. The mechanism is spectral: an eigenfunction φ_k with eigenvalue λ_k oscillates on the wavelength $\lambda_k^{-1/2}$, and Weyl's law $N(\lambda) \sim c\,\mathrm{Vol}(M)\lambda^{d_M/2}$ (d_M the intrinsic dimension) grows the count of modes below a fixed scale with resolution.

Lemma 1 (Fixed-band consistency)**.** *At a fixed low-mode budget m, as the sample grows the graph operator converges spectrally to the manifold operator [14], so the fixed low-λ band sharpens and geodesic fidelity* converges toward its continuum value *and is predicted to improve asymptotically (the spectral-convergence mechanism of Paper I [20]). Convergence does not force step-wise monotonicity in N; the monotone finite-sample trend below is the* registered empirical *result, not a consequence of the lemma.*

Holding instead the resolved scale τ fixed (spectral-gap matching), $m = N(\tau)$ grows with N; the newly resolved modes near τ have wavelength approaching the sampling scale and, for a non-ideal substrate, inject sub-geodesic detail, so geodesic fidelity is nonincreasing—Paper I's registered inversion [20]. We replicate it on a new substrate: 768-dimensional multilingual book-paragraph embeddings (Project Gutenberg translations; 32,000 points, three seeds). Fixed $m = 10$: geometry preservation rises with N (trend $+0.4, +1.0, +1.0$). Spectral-gap-matched, $m : 10 \to 121/126/159$: geometry *collapses* (trend -1.0 every seed); the budgets agree at the smallest N (gap ≤ 0.002) (*Replicated*). Opposite verdicts about the same data, set by the observer's rate: "clean geometry" is a relation between observer resolution and substrate scale, not a substrate invariant.

VIII. A Consumer-Relative Rate–Distortion Function

The consumer-projected distortion is the single-letter measure $d_{\mathcal{C}}(x, \hat{x}) = (x - \hat{x})^\top P_{\mathcal{C}}(x - \hat{x})$ for a *fixed* $P_{\mathcal{C}}$ (constant, or the average $\mathbb{E}_x[P_{\mathcal{C}}(x)]$ under the proviso of Prop. 1); define $R_{\mathcal{C}}(D) = \inf\{I(X; \hat{X}) : \mathbb{E}\, d_{\mathcal{C}}(X, \hat{X}) \leq D\}$. This fixed-$P_{\mathcal{C}}$ model must be kept distinct from the *local* observation metric $P_{\mathcal{C}}(x)$ of Prop. 1. For a point-dependent $P_{\mathcal{C}}(x)$ the operational rate–distortion theorem still *exists*—$d_{\mathcal{C}}$ is then a general measurable single-letter distortion, so Cor. 6 and Appendix B apply—but what is *lost* is the closed-form solution: there is no single *linear* tilted source (a direction invisible at one point can be visible at another), so the Berger reduction and reverse water-filling below are specific to the fixed weighted-quadratic model, whose theory is classical [4].

Proposition 6 (Gaussian rate–distortion; Berger reduction)**.** *For $X \sim \mathcal{N}(0, \Sigma_x)$, in the basis diagonalizing $P_{\mathcal{C}}^{1/2}\Sigma_x P_{\mathcal{C}}^{1/2}$ with eigenvalues γ_i,*

$$R_{\mathcal{C}}(D) = \sum_i \tfrac{1}{2}\log_2^+ \frac{\gamma_i}{\theta}, \qquad \sum_i \min(\gamma_i, \theta) = D,$$

a reverse water-filling on the $P_{\mathcal{C}}$-weighted source spectrum. For $P_{\mathcal{C}} = I$ this is Shannon's Gaussian $R(D)$ [2]; the estimation dual is Gauss–Markov least squares [3].

Proof. $\mathrm{tr}(P_{\mathcal{C}}\Sigma_\delta) = \|P_{\mathcal{C}}^{1/2}(x - \hat{x})\|^2$; with $y = P_{\mathcal{C}}^{1/2}x$ this is MSE on $y \sim \mathcal{N}(0, P_{\mathcal{C}}^{1/2}\Sigma_x P_{\mathcal{C}}^{1/2})$, and Shannon water-filling applies. The transform is invertible off $\ker P_{\mathcal{C}}$, on which the consumer is blind and no rate is spent. □

Bits water-fill on directions the consumer reads *and* the source varies, spending nothing on $\ker P_{\mathcal{C}}$—why a reconstruction-optimal code ($P_{\mathcal{C}} = I$ allocation) is beaten by a consumer-matched one when $P_{\mathcal{C}} \neq I$, and why the winner *flips* with $P_{\mathcal{C}}$ (§IV). The $H = I$ control confirms it: only at the isotropic $P_{\mathcal{C}} = I$ is reconstruction the right objective, and there it is exactly right.

Proposition 7 (Consumer-lossless floor; discrete source)**.** *Let X be discrete with finite entropy, Π the projector onto* $\mathrm{range}(P_{\mathcal{C}})$*. There exist codes with* $\mathrm{tr}(P_{\mathcal{C}}\Sigma_\delta) = 0$ *at rate*

$$R_{\mathcal{C}}(0) = H(\Pi X) \le H(X), \qquad H(X) - R_{\mathcal{C}}(0) = H(X \mid \Pi X) \ge 0,$$

with equality iff Π is injective on $\mathrm{supp}(X)$*; the saving is strict exactly when distinct symbols differ only within the consumer-invisible subspace* $\ker P_{\mathcal{C}}$*.*

Proof. Achievability: $\mathrm{tr}(P_{\mathcal{C}}\Sigma_\delta)$ depends on δ only through $\Pi\delta$; sending ΠX losslessly and reconstructing $(\mathrm{I} - \Pi)X$ by its conditional mode gives $\Pi\delta = 0$, zero distortion, at rate $H(\Pi X)$. *Converse:* zero distortion forces $\Pi\hat{X} = \Pi X$ a.s., so $I(X;\hat{X}) \ge I(\Pi X; \Pi\hat{X}) = H(\Pi X)$. Hence $R_{\mathcal{C}}(0) = H(\Pi X)$; if Π collapses two support points, $H(\Pi X) < H(X)$. ☐

Remark (Continuous sources). When the tilted source $Y = P_{\mathcal{C}}^{1/2}X$ has a nondegenerate continuous component, lossless-to-the-consumer coding costs infinite rate ($R_{\mathcal{C}}(D) \to \infty$ as $D \to 0$, consistent with Prop. 6); if X lies entirely in $\ker P_{\mathcal{C}}$ then Y is degenerate and the floor is zero. The surviving statement is that at *every* $D > 0$ the water-filling of Prop. 6, equivalently the reduction of Cor. 5, spends zero rate on $\ker P_{\mathcal{C}}$: the operational rate depends only on the r-dimensional tilted source. This lossless-to-the-consumer phenomenon is exactly coding for computing and graph entropy in the discrete world [8], [9], here given a geometric read operator.

These form a *conceptual axis*, not a formal nesting: Shannon's Gaussian Euclidean-MSE specialization ($P_{\mathcal{C}} = I$, the isotropic observer), consumer-relative rate–distortion ($P_{\mathcal{C}}$ a fixed geometric, *identifiable* operator, Props. 6,7), and the information bottleneck [12]—which optimizes a different object (a stochastic relevance variable) and to which the consumer-relative term connects only through the local-Gaussian relevance limit ($Y = \mathcal{C}(X)$ deterministic, second-order). We claim no containment theorem. The middle term is the operational one—$P_{\mathcal{C}}$ is PSD, water-fillable, and its read subspace is recoverable from consumer queries under the specified probing model (Prop. 4)—and its coding theorem (achievability and converse, fixed $P_{\mathcal{C}}$, general source) is Appendix A.

IX. Two coding-theoretic consequences

The reduction to the tilted source (Lemma 5) and the coding theorem (§VIII, Appendix A) have two short consequences that place the consumer-relative function inside standard information theory: a finite-blocklength law and a source–channel separation.

Proposition 8 (Finite-blocklength reduction)**.** *Fix $P_{\mathcal{C}}$ of rank r and an i.i.d. source with a finite-second-moment reference letter. Under the excess- or average-distortion criterion the finite-blocklength consumer coding problem for $d_{\mathcal{C}}$ is operationally identical,* at every blocklength n, *to the mean-squared-error problem for the r-dimensional tilted source $Y = P_{\mathcal{C}}^{1/2}X$ (Lemma 5): $R_{\mathcal{C}}(n,\epsilon,D) = R_Y(n,\epsilon,D)$ for every n,ϵ.*

Proof. $R_{\mathcal{C}} \le R_Y$: run any Y-code and output an arbitrary constant on $\ker P_{\mathcal{C}}$ (zero distortion there), matching rate and distortion sample-for-sample. $R_{\mathcal{C}} \ge R_Y$: an X-code's decoder emits at most 2^{nR} reconstructions, hence at most 2^{nR} points in Y-space. Given any X-encoder and its induced Y-space codebook, re-assigning each source block to its minimum-distance codeword in Y^n cannot increase distortion and is a function of Y^n alone; hence both the excess-distortion event and the average distortion depend on Y^n and the codebook only, and the minimal codebook size is the D-covering number of Y^n—the r-dimensional MSE problem. The encoder's side information on $\ker P_{\mathcal{C}}$—genuinely correlated with Y when Σ_x couples the blocks—is thus operationally inert: covering Y is insensitive to data held only at the encoder. So $R_{\mathcal{C}}(n,\epsilon,D) = R_Y(n,\epsilon,D)$ exactly at every n. ☐

Corollary 1 (Dispersion counts read dimensions)**.** *For a Gaussian source under quadratic distortion, in the excess-distortion regime where the Kostina–Verdú [26] normal approximation holds, the reduction gives*

$$R_{\mathcal{C}}(n,\epsilon,D) = R_Y(D) + \sqrt{V_Y(D)/n}\,Q^{-1}(\epsilon) + O\left(\tfrac{\log n}{n}\right),$$

with effective dimension $r_D := \#\{$active water-filling modes$\} \le r$ (equal to r as $D \to 0$), never the ambient d; the tilted-Gaussian dispersion V_Y is cited—the scalar quadratic-Gaussian case [26], [27], the parallel case following by per-mode additivity of the tilted information—not re-derived.

Remark. Verbatim transfer is for the fixed-$P_{\mathcal{C}}$ Gaussian-quadratic reduction. For the true nonlinear divergence (Theorem 4: output coding of $\mathcal{C}(X)$ under a non-difference D_Y) the finite-blocklength theory ports only under Linder–Zamir [25] non-difference regularity—we keep that hedge.

Proposition 9 (Observer separation)**.** *Let X be i.i.d. read by a fixed consumer with distortion $d_{\mathcal{C}}$ (reference letter of finite second moment), transmitted at ρ source symbols per channel use over a memoryless channel of capacity C_{ch} without feedback. The source is transmissible within consumer distortion D if $\rho R_{\mathcal{C}}(D) < C_{\mathrm{ch}}$, and only if $\rho R_{\mathcal{C}}(D) \le C_{\mathrm{ch}}$; the boundary $\rho R_{\mathcal{C}}(D) = C_{\mathrm{ch}}$ is transmissible only when the distortion–rate curve is approachable from below in rate.*

Proof. Take $\rho = 1$. *Forward:* for $R_{\mathcal{C}}(D) < R < C_{\text{ch}}$, concatenate a rate-$R$ consumer source code (Theorem 2) with a capacity-achieving channel code; on the error event (probability $P_e \to 0$) the distortion contribution is $O(\sqrt{P_e}) \to 0$ by the second-moment hypothesis, so $\limsup_n \mathbb{E} d_{\mathcal{C}} \le D$. *Converse:* $nR_{\mathcal{C}}(D) \le I(X^n; \hat{X}^n) \le nC_{\text{ch}}$ by the RD converse and data processing across the memoryless channel. At the boundary $\rho R_{\mathcal{C}}(D) = C_{\text{ch}}$ transmissibility needs the distortion–rate curve approachable from below in rate: if $R_{\mathcal{C}}$ is strictly decreasing at D, codes for $D + 1/k$ use rate $R_{\mathcal{C}}(D+1/k) < C_{\text{ch}}$ and give $\limsup$-distortion D; on a flat segment of $R_{\mathcal{C}}$ it is not guaranteed. The only property used is that $d_{\mathcal{C}}$ is an admissible distortion with a legitimate $R_{\mathcal{C}}$. □

Remark. The capacity C_{ch} here is an ordinary Shannon channel capacity; it is *not* the identification threshold of §VI, whose $\rho = 1$ marks when the target's standardized margin can no longer be separated from the maximum of $N-1$ distractor scores—an order-statistic limit with no channel and no code. The intrinsic consumer-relative *channel* capacity that would turn that threshold into an operational coding limit remains open.

X. Successive refinement across observers

The single-observer theory (§VIII) serves one consumer at a time. When two consumers read the *same* Gaussian source in two stages—a base layer judged by consumer 1's read operator P_1, the full description by consumer 2's P_2—the frontier flagged in §II ("another consumer may read those directions") becomes a coding question we answer completely: we determine the entire two-stage rate region (Theorem 1), of which successive refinability and the exact rate loss are the corner, and we extend it to k observers—the paper's read operator turned into rate.

Let $X \sim \mathcal{N}(0, \Sigma_x)$ on $\mathbb{R}^d$, $\Sigma_x \succ 0$, be i.i.d.; consumer $i \in \{1,2\}$ has $P_i \succeq 0$ and distortion $d_i(x,\hat{x}) = (x-\hat{x})^\top P_i (x-\hat{x})$, with single-stage rate–distortion function R_i. Throughout we assume the coding regime $0 < D_i < \operatorname{tr}(P_i \Sigma_x)$, so each optimum $\Sigma_i^\star$ (below) is positive definite and the distortion constraint binds. Logarithms are natural and rates in nats throughout this section, §IX, and Appendix C (Prop. 6 is stated in bits). A two-stage code sends nR_1 bits decoded to $\hat{X}_1^n$ for consumer 1 and $n(R_2 - R_1)$ further bits so both messages decode to $\hat{X}_2^n$ for consumer 2, with $\mathbb{E}\frac{1}{n}\sum_t d_i(X_t, \hat{X}_{i,t}) \le D_i$. The pair (D_1, D_2) is *successively refinable* if the corner $(R_1, R_2) = (R_1(D_1), R_2(D_2))$ is achievable. Write $\Sigma(\hat{X}) := \operatorname{Cov}(X - \mathbb{E}[X \mid \hat{X}]) = \mathbb{E}[\operatorname{Cov}(X \mid \hat{X})]$. The two-stage achievable region for memoryless sources with per-stage distortion measures is Rimoldi's [33]: (R_1, R_2) achievable iff there exist $\hat{X}_1, \hat{X}_2$ with $\mathbb{E} d_i(X, \hat{X}_i) \le D_i$, $R_1 \ge I(X; \hat{X}_1)$, $R_2 \ge I(X; \hat{X}_1, \hat{X}_2)$; for our abstract-alphabet, unbounded-distortion instance we invoke the abstract-source successive-refinement coding theorem [34] (which extends Rimoldi's finite-alphabet per-stage-measure region to standard-Borel sources under a finite-distortion reference letter $\hat{x}_0 = 0$, $\mathbb{E} d_i(X, 0)^2 < \infty$ for Gaussian X); only its *achievability* direction is used—the converses below are self-contained. Uniqueness of the max-det optimizer (Lemma 4) forces the per-stage distortions to bind at D_i, which pins the corner $(R_1(D_1), R_2(D_2))$; the region itself (Theorem 1) needs neither.

Lemma 2 (Conditional mean serves every observer)**.** *For any $P \succeq 0$ and estimator $g(\hat{X})$, $\mathbb{E}[(X-g)^\top P (X-g)] \ge \mathbb{E}[(X - \mathbb{E}[X \mid \hat{X}])^\top P (X - \mathbb{E}[X \mid \hat{X}])] = \operatorname{tr}(P\,\Sigma(\hat{X}))$.*

Proof. Split $X - g = (X - \mathbb{E}[X \mid \hat{X}]) + (\mathbb{E}[X \mid \hat{X}] - g)$; the cross term vanishes since $\mathbb{E}[X - \mathbb{E}[X \mid \hat{X}] \mid \hat{X}] = 0$, and the second term is a nonnegative P-quadratic. □

Lemma 3 (Information bound, arbitrary joints)**.** *For any joint law of $(X, \hat{X})$ with $X \sim \mathcal{N}(0, \Sigma_x)$, $I(X; \hat{X}) \ge \frac{1}{2}\log\det\Sigma_x - \frac{1}{2}\log\det\Sigma(\hat{X})$ and $\Sigma(\hat{X}) \preceq \Sigma_x$.*

Proof. $I = h(X) - h(X \mid \hat{X})$, and $h(X \mid \hat{X}) = h(X - \mathbb{E}[X \mid \hat{X}] \mid \hat{X}) \le h(X - \mathbb{E}[X \mid \hat{X}]) \le \frac{1}{2}\log[(2\pi e)^d \det\Sigma(\hat{X})]$ by conditioning and Gaussian entropy maximization. The covariance bound is $\Sigma(\hat{X}) = \Sigma_x - \operatorname{Cov}(\mathbb{E}[X \mid \hat{X}]) \preceq \Sigma_x$. □

Lemma 4 (Gaussian optimum: max-det form and uniqueness)**.** *For $\Sigma_x \succ 0$, $P \succeq 0$, $D > 0$, $R_P(D) = \frac{1}{2}\log\det\Sigma_x - \frac{1}{2}\log\det\Sigma^\star(P,D)$, where $\Sigma^\star(P,D) = \arg\max\{\log\det\Sigma : 0 \preceq \Sigma \preceq \Sigma_x,\ \operatorname{tr}(P\Sigma) \le D\}$, and the maximizer is unique.*

Proof. Converse: Lemmas 2–3 give, for any admissible channel, $I \ge \frac{1}{2}\log\det\Sigma_x - \frac{1}{2}\log\det\Sigma(\hat{X})$ with $\Sigma(\hat{X})$ feasible; minimizing the bound is the stated program. *Achievability:* the jointly Gaussian backward channel $X = \hat{X} + Z$, $Z \sim \mathcal{N}(0, \Sigma^\star) \perp \hat{X}$, $\operatorname{Cov}\hat{X} = \Sigma_x - \Sigma^\star \succeq 0$, attains it. *Uniqueness:* the feasible set is convex with an interior point; since $\log\det = -\infty$ on singular matrices the maximizer lies in $\mathbb{S}^d_{++}$, where $\log\det$ is strictly concave. □

Theorem 1 (Complete two-stage rate region)**.** *Let $\Sigma_i^\star := \Sigma^\star(P_i, D_i)$. With R_2 the total rate to both stages, the closure of the achievable two-stage rate region is*

$$\mathcal{R}(D_1, D_2) = \Big\{(R_1, R_2) :\ \exists\, \Sigma_f \preceq \Sigma_b \preceq \Sigma_x \text{ with } \operatorname{tr}(P_1\Sigma_b) \le D_1,\ \operatorname{tr}(P_2\Sigma_f) \le D_2,$$
$$R_1 \ge \tfrac{1}{2}\log\tfrac{\det\Sigma_x}{\det\Sigma_b},\quad R_2 \ge \tfrac{1}{2}\log\tfrac{\det\Sigma_x}{\det\Sigma_f}\Big\},$$

the base residual $\Sigma_b = \Sigma(\hat{X}_1)$ nesting above the pair residual $\Sigma_f = \Sigma(\hat{X}_1, \hat{X}_2)$.

Proof. Converse. For any two-stage code put $\Sigma_b := \Sigma(\hat{X}_1)$, $\Sigma_f := \Sigma(\hat{X}_1, \hat{X}_2)$. Lemma 2 with $g = \hat{X}_1$ on $\sigma(\hat{X}_1)$ gives $\operatorname{tr}(P_1\Sigma_b) \le \mathbb{E}d_1 \le D_1$; with $g = \hat{X}_2$ on $\sigma(\hat{X}_1, \hat{X}_2)$—legal because $\hat{X}_2$ is a function of both messages, so the *pair*-conditional mean serves consumer 2 at least as well as $\hat{X}_2$—$\operatorname{tr}(P_2\Sigma_f) \le \mathbb{E}d_2 \le D_2$. Conditioning the law of total covariance on $\hat{X}_1$ yields $\Sigma_b = \Sigma_f + \mathbb{E}\big[\operatorname{Cov}(\mathbb{E}[X \mid \hat{X}_1, \hat{X}_2] \mid \hat{X}_1)\big] \succeq \Sigma_f$, and Lemma 3 gives $\Sigma_b \preceq \Sigma_x$; so $\Sigma_f \preceq \Sigma_b \preceq \Sigma_x$. Finally Lemma 3 on $\hat{X}_1$ and on the pair gives $R_1 \ge I(X; \hat{X}_1) \ge \frac{1}{2}\log\det\Sigma_x/\det\Sigma_b$ and $R_2 \ge I(X; \hat{X}_1, \hat{X}_2) \ge \frac{1}{2}\log\det\Sigma_x/\det\Sigma_f$. *The converse uses neither a Markov-chain assumption nor max-det uniqueness (Lemma 4)—the nesting comes straight from the law of total covariance. Achievability.* Fix feasible $\Sigma_f \preceq \Sigma_b \preceq \Sigma_x$ and set $\Lambda_b := \Sigma_b^{-1} - \Sigma_x^{-1} \preceq \Lambda_f := \Sigma_f^{-1} - \Sigma_x^{-1}$ (antitone inversion). The independent-increment Gaussian observations $Z_1 := \Lambda_b^{1/2}X + M_1$ and $Z_2 := (\Lambda_f - \Lambda_b)^{1/2}X + M_2$, $M_1, M_2 \sim \mathcal{N}(0, I)$ independent, have additive Fisher information, so $\operatorname{Cov}(X \mid Z_1) = (\Sigma_x^{-1} + \Lambda_b)^{-1} = \Sigma_b$ and $\operatorname{Cov}(X \mid Z_1, Z_2) = \Sigma_f$; with $\hat{X}_1 = \mathbb{E}[X \mid Z_1]$, $\hat{X}_2 = \mathbb{E}[X \mid Z_1, Z_2]$ this attains the boundary rates, and the abstract-source achievability of [34] (finite alphabets: [33]) lifts the single-letter point to $\mathcal{R}$. Upward closure supplies the rest. ☐

The corner of $\mathcal{R}$ recovers—and simplifies the proof of—the two headline statements; uniqueness (Lemma 4) re-enters only there.

Corollary 2 (Refinability)**.** *(D_1, D_2) is successively refinable, i.e. the corner $(R_1(D_1), R_2(D_2)) \in \mathcal{R}$, if and only if the observers' optimal uncertainty ellipsoids nest, $\Sigma_2^\star \preceq \Sigma_1^\star$.*

Proof. The corner requires a feasible pair with $\det\Sigma_b \ge \det\Sigma_1^\star$ and $\det\Sigma_f \ge \det\Sigma_2^\star$; but Σ_b, Σ_f are feasible for the respective max-det programs, so uniqueness (Lemma 4) forces $\Sigma_b = \Sigma_1^\star$, $\Sigma_f = \Sigma_2^\star$. The region's nesting $\Sigma_f \preceq \Sigma_b$ is then exactly $\Sigma_2^\star \preceq \Sigma_1^\star$. ☐

Corollary 3 (Exact rate loss)**.** *Among two-stage codes whose base is stage-1-optimal ($R_1 = R_1(D_1)$, $\mathbb{E}d_1 \le D_1$) and whose second stage meets D_2, the minimal total rate is $\frac{1}{2}\log\det\Sigma_x - \frac{1}{2}\log\det\Sigma^\circ$, where $\Sigma^\circ := \arg\max\{\log\det\Sigma : \Sigma \preceq \Sigma_1^\star,\ \operatorname{tr}(P_2\Sigma) \le D_2\}$—the* same *reverse water-filling as $\Sigma^\star$ with the cap Σ_x replaced by $\Sigma_1^\star$, i.e. $\Sigma^\circ = \Sigma^\star(P_2, D_2; \Sigma_1^\star)$. Hence the rate loss is $L(D_1, D_2) = \frac{1}{2}\log(\det\Sigma_2^\star/\det\Sigma^\circ) \ge 0$, with $L = 0$ iff $\Sigma_2^\star \preceq \Sigma_1^\star$.*

Proof. Restrict $\mathcal{R}$ to $\Sigma_b = \Sigma_1^\star$ (Corollary 2) and minimize $R_2 = \frac{1}{2}\log\det\Sigma_x/\det\Sigma_f$ over feasible fine layers $\Sigma_f \preceq \Sigma_1^\star$, $\operatorname{tr}(P_2\Sigma_f) \le D_2$; maximizing $\det\Sigma_f$ is the stated constrained max-det program. $L \ge 0$ because Σ° is feasible for the looser $\Sigma_2^\star$ program, so $\det\Sigma^\circ \le \det\Sigma_2^\star$; $L = 0$ iff $\Sigma_2^\star$ already nests under $\Sigma_1^\star$ (then $\Sigma^\circ = \Sigma_2^\star$). ☐

The same region argument closes the k-observer case, settling a frontier the corner theorem left open.

Corollary 4 (The k-observer chain)**.** *For read operators $P_1, \dots, P_k \succeq 0$ with targets $D_1, \dots, D_k$, write T_i for the* cumulative *rate decoded through stage i (so $T_2 = R_2$ of Theorem 1). The achievable region is $\{(T_1, \dots, T_k) : \exists\ \Sigma_k \preceq \cdots \preceq \Sigma_1 \preceq \Sigma_x,\ \operatorname{tr}(P_i\Sigma_i) \le D_i,\ T_i \ge \frac{1}{2}\log\det\Sigma_x/\det\Sigma_i\ \forall i\}$, and $(D_1, \dots, D_k)$ is refinable iff the k single-stage optima form a nested chain $\Sigma_k^\star \preceq \cdots \preceq \Sigma_1^\star \preceq \Sigma_x$.*

Proof. Put $\Sigma_i := \Sigma(\hat{X}_{1:i})$: Lemma 2 gives each trace bound, the law of total covariance (conditioned on $\hat{X}_{1:i-1}$) the chain nesting, and Lemma 3 the cumulative-rate bounds. A feasible chain is realized by independent Fisher increments $Z_i = \Delta_i^{1/2}X + M_i$, $\Delta_i := \Lambda_i - \Lambda_{i-1} \succeq 0$ ($\Lambda_i := \Sigma_i^{-1} - \Sigma_x^{-1}$, $\Lambda_0 := 0$, M_i i.i.d.), so $\operatorname{Cov}(X \mid Z_{1:i}) = \Sigma_i$. The two-stage abstract-source arguments of [34] extend verbatim to k stages (cf. the discrete multiresolution region [29]), and for the present Gaussian instance the independent-increment construction with the covering argument of Prop. 8 yields k-stage achievability directly. Refinability pins each $\Sigma_i = \Sigma_i^\star$ by uniqueness, and the chain nesting reads $\Sigma_i^\star \preceq \Sigma_{i-1}^\star$. ☐

Remark. The two-stage backward channel does *not* iterate for $k \ge 3$: one contraction fixes the Fisher information only up to a rotation, so re-using it misplaces information at the next stage. The independent-increment (equivalently physically-degraded Gaussian) realization above is the correct k-stage construction; the $k = 2$ map is used exactly once and is unaffected.

Three consequences make the characterization legible. *(i) Commuting observers.* If Σ_x, P_1, P_2 are simultaneously diagonalizable with mode variances σ_k^2 and weights $p_k^{(i)}$, then $e_k^{(i)} = \min(\sigma_k^2, \theta_i/p_k^{(i)})$ (water levels θ_i from $\sum_k p_k^{(i)}e_k^{(i)} = D_i$), and refinability is $e_k^{(2)} \le e_k^{(1)}$ for every mode. *(ii) Equitz–Cover, recovered.* For $P_1 = P_2 = P$ (not necessarily commuting with Σ_x), whiten by $\tilde{P} = \Sigma_x^{1/2}P\Sigma_x^{1/2} = V\operatorname{diag}(\tilde{p}_k)V^\top$: then $\Sigma^\star(D) = \Sigma_x^{1/2}V\operatorname{diag}(\min(1, \theta_D/\tilde{p}_k))V^\top\Sigma_x^{1/2}$ is Loewner-monotone in D in a *D-independent* eigenbasis, so $D_2 \le D_1$ gives $\Sigma_2^\star \preceq \Sigma_1^\star$ automatically—a Gaussian source is successively refinable under any single weighted-quadratic measure, the $P = I$ case being Equitz–Cover [32]; conflict requires genuinely distinct observers. *(iii) Kernel overhead.* If the base layer codes a direction in $\ker P_2$—one consumer 2 cannot read ($p_k^{(1)} > 0$, $e_k^{(1)} < \sigma_k^2$, $p_k^{(2)} = 0$)—then $e_k^{(2)} = \sigma_k^2 > e_k^{(1)}$ and refinability fails strictly: rate spent on directions the next observer cannot read is unrecoverable. Concretely, for $\Sigma_x = I_2$, $P_1 = \operatorname{diag}(1, 0)$, $P_2 = \operatorname{diag}(0, 1)$, the optima $\Sigma_1^\star = \operatorname{diag}(D_1, 1)$ and $\Sigma_2^\star = \operatorname{diag}(1, D_2)$ never nest and $\Sigma^\circ = \operatorname{diag}(D_1, D_2)$, so $L = \frac{1}{2}\log(1/D_1) = R_1(D_1)$: the minimal total rate is $R_1(D_1) + R_2(D_2)$, zero reuse. This is "one observer's noise is another's signal" (§II) as an exact coding theorem.

We confirmed Theorem 1 and Corollaries 2–4 numerically (committed with the ledger [22]): the region converse survives 5,000+ random admissible codes, the achievability construction reproduces every feasible (Σ_b, Σ_f) and the $k = 3$ chain, and the corner recovers the nesting characterization, the Equitz–Cover recovery, and the orthogonal-observer loss $L = R_1(D_1)$; sealed preregs GO-P-2026-022 (corner) and GO-P-2026-023 (region and k-observer), each with a fresh-context derivation-grade pass logged—the latter caught that the two-stage backward channel does not iterate for $k \geq 3$.

Related work. The successive-refinement line is Koshelev [28], Equitz–Cover [32], Rimoldi [33] (per-stage measures may differ), and, for abstract sources, Effros [29] and Kostina–Tuncel [34], which supply the operational region invoked above. Two works are closest, and neither subsumes ours. Nayak *et al.* [30] treat one vector Gaussian source under *individual* (per-component) MSE criteria—the same coordinate constraints at both stages—and already show that vector-Gaussian refinability can fail; Op 't Veld and Gastpar [31] refine a rank-deficient Gaussian *projection* to full reconstruction ($P_2 = I$), exhibiting the kernel-overhead rate penalty. Our contribution is neither of those qualitative phenomena—both are known—but their common generalization: the complete rate region (Thm. 1) for two *arbitrary, possibly non-commuting, possibly different-kernel* positive-semidefinite read operators, whose corner is the single Loewner condition $\Sigma_2^\star \preceq \Sigma_1^\star$ (Cor. 2) with the closed-form max-det rate loss (Cor. 3), extended to the k-observer chain (Cor. 4); to our knowledge this general two-operator statement has not appeared explicitly. The covariance-distortion-constraint and side-informed vector-Gaussian successive-refinement lines [35], [36] impose *matrix* distortion constraints or add side information rather than two scalar weighted-quadratic criteria, and do not specialize to the $\Sigma_2^\star \preceq \Sigma_1^\star$ corner with the closed-form max-det loss. Open beyond it: non-Gaussian sources.

XI. A Boundary: The Refuted Density Quotient (GO-5)

Not every plausible face survives a sealed prediction. In the spectral setting the $\alpha{=}1$ anisotropic normalization of a diffusion operator cancels sampling density and recovers density-free geometry [13]: there the density mode lies in $\ker P_{\mathcal{C}}$. We conjectured this density quotient likewise restores the invariant in a *non-spectral* domain (locally-scaled ADC retrieval, a hubness correction [15]). It does not, across four prospective tests (direct affinity twice, ADC on real embeddings, non-spectral diffusion-distance). The reason is Proposition 6: for a direct inner-product consumer the density is *read*—it is not in $\ker P_{\mathcal{C}}$—so quotienting it removes signal and manufactures anti-hubs; in the diffusion domain the direction is right ($\alpha{=}1$ optimal, recovered non-spectrally) but the effect is under 2% and *not density-specific* (a uniform-density control gains as much). Per its sealed registration the conjecture is *Refuted*. The sharpened statement—the $\alpha{=}1$ cancellation is a property of the diffusion operator and does not transfer to direct observation—is Proposition 6 read backwards: rate is saved on $\ker P_{\mathcal{C}}$ only when the nuisance actually lies there. The refutation sharpens the theory rather than undermining it.

XII. Negative Results and Methodology

Twelve standing negatives bound the theory and are reported with the prominence of the positives. Reconstruction cosine is not a proxy for key quality (Paper II [21], the motivating negative). The flip is observable only for reconstruction-*matched* codes: absent a tie one code Pareto-dominates and reconstruction is not falsified (a prospective miss locating a precondition, consistent with the water-filling of Prop. 6). The projected-variance trace governs the discriminating comparison but is not a complete rank statistic over all codes. The density quotient is treated in §XI. Two negatives are methodological self-corrections: in two instances a substantive claim passed at $12/12$ while an over-strict secondary gate we had specified failed, and in two others a degenerate substrate was detected only by a regime-coverage gate.

Every empirical claim was governed by a dated, hashed registry entry committed—sealed—before the first measurement it governs; bars were fixed ex ante; misses enter the ledger as misses; amendments only dated and logged before unblinding; prospective claims (§V,VII,XI) were blinded where feasible. The house rule is that the paper may not assert what the ledger cannot show.

XIII. Discussion

Table III summarizes the falsifiable core. In summary, Observation Theory identifies operational geometry with the pullback geometry of distinguishability and makes that geometry recoverable and experimentally consequential. It implies a change of default: where a conventional pipeline minimizes reconstruction and presumes that the task follows, the theory instead prescribes recovering the consumer's read operator (§V), water-filling bits on $P_{\mathcal{C}}$ rather than on I (§VIII), respecting the channel's vacuity threshold (§VI), and recognizing that the verdict is relative to the observation budget (§VII). Against the literatures it unifies—indirect source coding, coding for computing, rate–distortion–perception, task-oriented communication, loss-aware compression—the contribution is making the task's metric a single identifiable geometric object and testing it adversarially. The decisive validation is not conceptual but empirical, and we conducted it on a real model. A first preregistered blind probe on a Llama-3.2-3B attention layer (Gate B, GO-P-2026-020) *missed* its sealed bar: recovery was above chance but sub-threshold (0.567), and its registered quantizer pair was *not* reconstruction-matched, so reconstruction predicted the winner for free ($16/16$) and the projected predictor could not *beat* it—the same precondition that governs the synthetic dissociation. The corrected rematch (GO-P-2026-021) fixes exactly this: two key-error arms tied in reconstruction to $7.5{\times}10^{-9}$ but differing downstream (median relative divergence gap 1.85), with an improved probe. On the trained model it *clears all four sealed gates*—the blind $\mathrm{tr}(\hat{P}_{\mathcal{C}}\Sigma_\delta)$ predicts the reconstruction-invisible worse arm on all 16 instances (layers $\{8, 16\}$, 8 KV heads each) while reconstruction, tied to 7.5×10^{-9}, has no discriminative power: its residual sign is set by sub-nanoscale rounding

TABLE III
THE FALSIFIABLE CORE. EACH ROW IS A SEALED, BEFORE-THE-FACT REGISTRATION WITH AN EX-ANTE ACCEPTANCE BAR; FOUR ARE BORNE OUT AND ONE IS REFUTED AND CARRIED AS A BOUNDARY. A PROSPECTIVE TRAINED-MODEL (LLAMA-3.2-3B) HARDENING OF GO-2 FIRST MISSED ON A NON-RECONSTRUCTION-MATCHED DESIGN AND IS THEN CLEARED BY A CORRECTED REMATCH (§XIII).

Claim	**Prediction**	**Class**	**Result**
GO-1 identifiability	blind probe recovers the read subspace	predicted	overlap 0.94 vs 0.06 chance; predicts the flip 12/12
GO-2 distortion	a reconstruction-identical code flips its verdict with the consumer	demonstrated, replicated	flips on attention, retrieval, and optimization
GO-3 capacity	extreme-value vacuity threshold locates retrieval death	demonstrated	orders 14 corpora at Spearman $r_s = 0.991$
GO-4 rate	a fixed-budget verdict inverts under budget matching	replicated	inverts on real 768-d embeddings
GO-5 density quotient	α=1 restores the invariant non-spectrally	refuted	4 prospective misses; boundary

and here points the wrong way (2/16; not chance, an artifact of the exactly-tied construction), so the projected predictor strictly beats it ($16 > 2$). The read *subspace* is recovered at median overlap 0.647, below the synthetic 0.936 of §V, as expected for a real point-dependent $P_{\mathcal{C}}(x)$: the probing distribution differs from the activation distribution, and the softmax output metric is rank-deficient. This is worse-arm discrimination under one consumer, not a two-consumer verdict inversion; the theory's mechanism nonetheless operates on a trained Llama attention layer, called *before* compression by a blind probe, with the GO-P-2026-020 miss retained as an honest negative on the code design it exposed. Open theory: we settle the fixed-$P_{\mathcal{C}}$ achievability and converse of $R_{\mathcal{C}}(D)$ in Appendix A and lift it to a fully point-dependent *surrogate* metric (Cor. 6), and Appendix B settles the *true* nonlinear divergence too—its single-letter theorem at every D (Thm. 3) and the fact that the true nonlinear problem reduces *exactly* to rate–distortion coding of the reachable output source $\mathcal{C}(X)$ (Thm. 4), with the quotient floor unconditional (Cor. 7). What remains open is narrower: the curved-image high-resolution case, the multi-observer rate region *under the true divergence* (distinct quotients, no common Gaussian reduction; the Gaussian weighted-quadratic region is settled in §X), the consumer-relative *channel* coding theorem behind §VI, and whether the flip is exactly $R_{\mathcal{C}}$'s argmin migrating with $P_{\mathcal{C}}$.

The coding-theoretic core of these claims lives in the appendices, and a reader focused on the information theory should read them next. Appendix A proves the achievability and converse of the consumer-relative rate–distortion function $R_{\mathcal{C}}(D)$ for a general source; Appendix B then extends the theorem to the *true* nonlinear consumer divergence and shows that the problem reduces *exactly* to ordinary source coding of the consumer output (the read-operator "tilt" being $\mathcal{C}$ itself). These are the results that make "distortion is what the consumer reads" a coding theorem rather than a heuristic, and they are load-bearing for every rate claim in the body.

Underlying all of this is a single principle: an object has no unique operational geometry; each observer $\mathcal{O} = (\mathcal{C}, G, B)$ pulls back one, and the quotient $X/{\sim_{\mathcal{C}}}$ is the domain on which it acts. Four of the five registered predictions are confirmed, one is refuted, and twelve standing negatives are retained; the framework is now positioned within the established literature.

APPENDIX A
THE CONSUMER-RELATIVE CODING THEOREM

Throughout the main theorem, $P_{\mathcal{C}}$ is a *fixed* PSD operator (constant, or $\mathbb{E}_x[P_{\mathcal{C}}(x)]$ under the proviso of Prop. 1); Cor. 6 then lifts the operational statement to a point-dependent $P_{\mathcal{C}}(x)$ surrogate metric.

Lemma 5 (Reduction to the tilted source)**.** *Let $Y = P_{\mathcal{C}}^{1/2}X$, on $\operatorname{range}(P_{\mathcal{C}})$ of dimension $r = \operatorname{rank} P_{\mathcal{C}}$. Then $d_{\mathcal{C}}(x, \hat{x}) = \|y - \hat{y}\|^2$ with $\hat{y} = P_{\mathcal{C}}^{1/2}\hat{x}$, and $R_{\mathcal{C}}(D) = R_Y(D)$, the MSE rate–distortion function of Y; $\ker P_{\mathcal{C}}$ costs zero rate.*

Proof. $d_{\mathcal{C}} = \|P_{\mathcal{C}}^{1/2}(x - \hat{x})\|^2 = \|y - \hat{y}\|^2$. Objective and constraint depend on $\hat{X}$ only through $\hat{Y}$, so by data processing $I(X; \hat{X}) \geq I(Y; \hat{Y})$; the minimizer places all information in Y (a channel $p(\hat{y} \mid y)$, then $\hat{X} = P_{\mathcal{C}}^{-1/2}\hat{Y}$ on the range, conditional mean on $\ker P_{\mathcal{C}}$), attaining equality. Hence the minimum is $R_Y(D)$. □

Theorem 2 (Achievability and converse)**.** *Let X take values in a standard Borel space, $\{X_t\}$ i.i.d. $\sim p_X$, with a measurable reconstruction alphabet and a reference letter $\hat{x}_0$ satisfying $\mathbb{E}\, d_{\mathcal{C}}(X, \hat{x}_0) < \infty$ and $\mathbb{E}\, d_{\mathcal{C}}(X, \hat{x}_0)^2 < \infty$. For every $D > 0$:* (i) *for any $R > R_{\mathcal{C}}(D)$ there are $(2^{nR}, n)$ codes with $\limsup_n \mathbb{E}\, \frac{1}{n} \sum_t d_{\mathcal{C}}(X_t, \hat{X}_t) \leq D$;* (ii) *every code of expected distortion $\leq D$ has rate $\geq R_{\mathcal{C}}(D)$; and $R_{\mathcal{C}}$ is convex and nonincreasing.*

Proof. This is the single-letter rate–distortion theorem for a memoryless source on a standard Borel space with a measurable difference distortion and finite reference-letter moment [4]; we recall it in *information-density* form, which uses no differential entropies and so holds on general alphabets. *Achievability.* Fix $\varepsilon > 0$ and a test channel $p^*(\hat{x} \mid x)$ with $\mathbb{E}\, d_{\mathcal{C}} \le D$ and $I(X;\hat{X}) \le R_{\mathcal{C}}(D)+\varepsilon$; let $p_{\hat{X}}$ be the output marginal and $\imath(x;\hat{x}) = \log \frac{dP_{X\hat{X}}}{d(P_X \times P_{\hat{X}})}$ the information density. Draw $\{\hat{X}^n(w)\}_{w=1}^{2^{nR}}$ i.i.d. $\sim \prod_t p_{\hat{X}}$; encode x^n by any w with $\frac{1}{n}\sum_t \imath(x_t;\hat{X}_t(w)) \le I(X;\hat{X}) + \varepsilon$ *and* $\frac{1}{n}\sum_t d_{\mathcal{C}}(x_t,\hat{X}_t(w)) \le D+\varepsilon$, else declare failure and reconstruct with the constant $\hat{x}_0^n$. Let $\mathcal{T}$ be the joint event $\{\frac{1}{n}\sum_t \imath \le I+\varepsilon,\ \frac{1}{n}\sum_t d_{\mathcal{C}} \le D+\varepsilon\}$; by the weak law $P^n_{X\hat{X}}(\mathcal{T}) \to 1$, so for all but an $o(1)$-probability set $\mathcal{G}_n$ of source sequences the *conditional* probability $P_{\hat{X}^n|X^n=x^n}(\mathcal{T}) \ge 1 - o(1)$. Changing measure from $P_{\hat{X}^n|x^n}$ to the codebook marginal $P_{\hat{X}^n} = \prod_t p_{\hat{X}}$ on $\mathcal{T}$ costs at most $2^{n(I+\varepsilon)}$ (there the density $\prod_t \frac{dP_{\hat{X}|X}}{dP_{\hat{X}}} = 2^{\sum_t \imath} \le 2^{n(I+\varepsilon)}$), so for each $x^n \in \mathcal{G}_n$ a single random codeword passes the encoding test with conditional probability $\ge 2^{-n(I+2\varepsilon)}(1-o(1))$. Hence $\Pr[\text{fail} \mid x^n] \le \big(1 - 2^{-n(I+2\varepsilon)}(1-o(1))\big)^{2^{nR}} \to 0$ for $R > I + 2\varepsilon$, uniformly over $x^n \in \mathcal{G}_n$, and $\Pr[\text{fail}] \le \Pr[\mathcal{G}_n^c] + \sup_{x^n \in \mathcal{G}_n} \Pr[\text{fail} \mid x^n] \to 0$. On success the distortion is $\le D+\varepsilon$; on failure it is $\frac{1}{n}\sum_t d_{\mathcal{C}}(X_t,\hat{x}_0)$, whose contribution to the expectation is $\le \sqrt{\mathbb{E}[(\frac{1}{n}\sum_t d_{\mathcal{C}}(X_t,\hat{x}_0))^2]}\,\sqrt{\Pr[\text{fail}]} \to 0$ by Cauchy–Schwarz, finite by the second-moment hypothesis on $\hat{x}_0$. Thus $\limsup_n \mathbb{E}\,\frac{1}{n}\sum_t d_{\mathcal{C}} \le D+\varepsilon$; let $\varepsilon \to 0$. *Converse.* With $D_t = \mathbb{E}\, d_{\mathcal{C}}(X_t,\hat{X}_t)$, $nR \ge H(W) \ge I(X^n;\hat{X}^n) \ge \sum_t I(X_t;\hat{X}_t) \ge \sum_t R_{\mathcal{C}}(D_t) \ge nR_{\mathcal{C}}(\frac{1}{n}\sum_t D_t) \ge nR_{\mathcal{C}}(D)$, by memorylessness, the definition of $R_{\mathcal{C}}$, Jensen (convexity), and monotonicity. Convexity and monotonicity follow from time-sharing and distortion relaxation. □

Corollary 5. *$R_{\mathcal{C}}(D)$ equals the MSE rate–distortion of the r-dimensional tilted source $Y = P_{\mathcal{C}}^{1/2}X$; for Gaussian X it is Prop. 6; discretely, $R_{\mathcal{C}}(0) = H(\Pi X)$ (Prop. 7). The operational rate to serve a consumer is governed entirely by the source projected into the read subspace—the precise sense in which Shannon's theorem is the corner $P_{\mathcal{C}} = I$, where the read subspace is everything.*

Corollary 6 (Point-dependent surrogate)**.** *The coding theorem holds verbatim for the* point-dependent *quadratic distortion $d_{P_{\mathcal{C}}}(x,\hat{x}) = (\hat{x}-x)^\top P_{\mathcal{C}}(x)(\hat{x}-x)$ whenever $P_{\mathcal{C}}(\cdot)$ is measurable with $\|P_{\mathcal{C}}(x)\| \le \Lambda < \infty$ and $\mathbb{E}\|X\|^4 < \infty$. The achievability and converse of Thm. 2 nowhere use that the distortion is a* difference *distortion—only that it is a nonnegative measurable single-letter distortion with a finite second-moment reference letter, and $d_{P_{\mathcal{C}}}(x,\hat{x}_0) \le \Lambda\|\hat{x}_0 - x\|^2$ supplies both. Hence the operational rate–distortion function $R_{\mathcal{C}}(D) = \inf\{I(X;\hat{X}) : \mathbb{E}\, d_{P_{\mathcal{C}}}(X,\hat{X}) \le D\}$ exists and is achievable for a fully point-dependent read operator.*

What the fixed-$P_{\mathcal{C}}$ specialization keeps and the point-dependent case loses is therefore sharp: *(i)* the closed-form reverse water-filling (Cor. 5) needs a single *linear* tilt $Y = P_{\mathcal{C}}^{1/2}X$; a point-dependent $P_{\mathcal{C}}(x)$ admits *no* tilted-source representation at all—the map $x \mapsto P_{\mathcal{C}}(x)^{1/2}x$ does not linearize the distortion, since in general $P_{\mathcal{C}}(x)^{1/2}(\hat{x}-x) \ne P_{\mathcal{C}}(\hat{x})^{1/2}\hat{x} - P_{\mathcal{C}}(x)^{1/2}x$—so $R_{\mathcal{C}}$ has no closed form, even though it exists as the rate–distortion function of a general measurable single-letter distortion (Cor. 6); *(ii)* the exact floor $R_{\mathcal{C}}(0) = H(X/\!\sim_{\mathcal{C}})$ requires the integrability/compatibility conditions of §II to identify the quotient; and *(iii)* $d_{P_{\mathcal{C}}}$ is the *leading-order surrogate* for the true divergence $D_Y(\mathcal{C}(x),\mathcal{C}(\hat{x}))$, so coding to it certifies the true downstream distortion only in the high-rate (small-D) regime. The genuinely open problem is thus not the existence of $R_{\mathcal{C}}(D)$ under a point-dependent metric—Cor. 6 settles that—and, as Appendix B shows, not the true nonlinear divergence either: Thm. 3 gives its single-letter theorem at every D, and Thm. 4 its exact reduction (the tilt is $\mathcal{C}$ itself, with no linear surrogate needed). What remains open is narrower, and stated in Appendix B, §B-D.

Appendix B
The Coding Theorem for the True Nonlinear Divergence

Here X is i.i.d. $\sim p_X$ on a Polish space $\mathcal{X}$, reconstruction alphabet $\hat{\mathcal{X}} = \mathcal{X}$, $\mathcal{C} : \mathcal{X} \to \mathcal{U}$ Borel into a Polish output space, and $D_Y : \mathcal{U} \times \mathcal{U} \to [0,\infty]$ jointly Borel with $D_Y(u,u) = 0$. The *true distortion* is $d_{\mathcal{C}}(x,\hat{x}) := D_Y(\mathcal{C}(x),\mathcal{C}(\hat{x}))$ and $R_{\mathcal{C}}^{\text{true}}(D) := \inf\{I(X;\hat{X}) : \mathbb{E}\, d_{\mathcal{C}}(X,\hat{X}) \le D\}$. The three results settle, in turn, existence of the single-letter theorem at non-vanishing D, the exact nonlinear tilt, and the quantitative relation to the quadratic surrogate $d_{P_{\mathcal{C}}}$ of Cor. 6. The load-bearing identities (Thm. 4, Cor. 7) and the high-resolution formula (Thm. 5) were confirmed to machine precision by an independent Blahut–Arimoto / reverse-water-filling check committed with the ledger [22].

A. Existence: the theorem ports verbatim

Theorem 3 (Single-letter theorem for the true divergence)**.** *If there is $\hat{x}_0$ with $\mathbb{E}\, d_{\mathcal{C}}(X,\hat{x}_0)^2 < \infty$, then for every $D > 0$, $R_{\mathcal{C}}^{\text{true}}(D)$ is the operational rate–distortion function (achievability for any $R > R_{\mathcal{C}}^{\text{true}}(D)$; converse), and $R_{\mathcal{C}}^{\text{true}}$ is convex and nonincreasing.*

Proof. $d_{\mathcal{C}}$ is a nonnegative, jointly Borel, single-letter distortion (a composition of Borel maps). The information-density achievability and converse of Thm. 2 use only nonnegativity, joint measurability, memorylessness, and the reference-letter second moment; they nowhere use symmetry, the triangle inequality, difference structure, or $d(x,\hat{x}) = 0 \iff x = \hat{x}$—as Cor. 6 records. The proof applies verbatim with $d_{\mathcal{C}}$ for $d_{P_{\mathcal{C}}}$ [23]. Nothing about the nonlinearity of $D_Y \circ (\mathcal{C},\mathcal{C})$ obstructs the classical theorem; the existence half of the "open problem" was mislocated. □

B. The exact nonlinear tilt: reduction to the output source

Theorem 4 (Output reduction; the tilt is $\mathcal{C}$)**.** *Let $U := \mathcal{C}(X)$, $V := \mathcal{C}(\hat{\mathcal{X}}) \subseteq \mathcal{U}$, and $R_U(D) := \inf\{I(U;\hat{U}) : \hat{U} \in V$ a.s., $\mathbb{E}\, D_Y(U,\hat{U}) \le D\}$ the ordinary rate–distortion function of the output source U under D_Y, reconstruction alphabet V. With D_Y jointly Borel and a universally measurable right inverse $s : V \to \hat{\mathcal{X}}$ of $\mathcal{C}$ (automatic for Borel $\mathcal{C}$ between Polish spaces: V is analytic and Jankov–von Neumann uniformization supplies s [24]), $R^{\text{true}}_{\mathcal{C}}(D) = R_U(D)$ for every $D \ge 0$.*

Proof. ($\ge$) Any admissible $\hat{X}$ gives $\hat{U} := \mathcal{C}(\hat{X}) \in V$ with $\mathbb{E}D_Y(U,\hat{U}) = \mathbb{E}d_{\mathcal{C}} \le D$, and $I(X;\hat{X}) \ge I(U;\hat{X}) \ge I(U;\hat{U})$ by data processing (U a function of X; $\hat{U}$ a function of $\hat{X}$). ($\le$) Given a test channel $p(\hat{u} \mid u)$ with $I(U;\hat{U}) \le R_U(D) + \varepsilon$, draw $\hat{U} \mid X \sim p(\cdot \mid \mathcal{C}(X))$ and set $\hat{X} := s(\hat{U})$; then $\mathcal{C}(\hat{X}) = \hat{U}$, so $\mathbb{E}d_{\mathcal{C}} = \mathbb{E}D_Y(U,\hat{U}) \le D$, and $X \to U \to \hat{U} \to \hat{X}$ is Markov, giving $I(X;\hat{X}) \le I(X;\hat{U}) = I(U;\hat{U})$. The equality uses *both* $I(U;\hat{U} \mid X) = 0$ (U is a deterministic function of X) and $I(X;\hat{U} \mid U) = 0$ (the Markov property), so $I(X;\hat{U}) = I(X,U;\hat{U}) = I(U;\hat{U})$. The universally measurable s defines the joint law of $(X,\hat{X})$; on the standard Borel product this disintegrates into a *Borel* conditional kernel, so the exhibited channel is feasible (universal measurability is used only to build the joint law). Because $d_{\mathcal{C}}(x,\hat{x})$ depends on $\hat{x}$ only through $\mathcal{C}(\hat{x}) \in V$, restricting reconstructions to $\{s(\hat{u})\}$ loses nothing—the reduction is tight. Let $\varepsilon \to 0$. □

Corollary 7 (Quotient floor)**.** *If D_Y is a* strict *divergence ($D_Y(u,u') = 0 \iff u = u'$) and X is discrete with finite entropy, then $R^{\text{true}}_{\mathcal{C}}(0) = H(\mathcal{C}(X)) = H(X/\!\sim_{\mathcal{C}})$. These two hypotheses are the only ones:* none *of the integrability, constant-rank, or leaf-compatibility conditions the* surrogate *needs (§II, Cor. 6) arises, because the true divergence defines the quotient globally rather than infinitesimally. (If D_Y is degenerate, replace $\mathcal{C}(X)$ by its quotient under D_Y-indistinguishability.) Prop. 7 is the linear case $\mathcal{C} = \Pi$.*

Corollary 8 (Order of operations)**.** *The reduction of Thm. 4 removes all dependence on $\ker P_{\mathcal{C}}$, quotienting it* exactly *prior to any linearization; the surrogate's pathologies (singular weights, non-integrable null distributions) are artifacts of linearizing before quotienting. The "nonlinear tilt" has a closed-form* reduction*—it is $\mathcal{C}$ itself; what has no closed form is a tilt constrained linear in x, which the true problem never needed. Residual regularity is then governed not by a $\ker P_{\mathcal{C}}$ singularity but by the output-side conditions of Thm. 5.*

Remark (Provenance). Thm. 4 is elementary—the deterministic, two-sided cousin of the indirect problem of Dobrushin–Tsybakov, Wolf–Ziv, and Witsenhausen—and we claim structural, not technical, novelty. The constraint $\hat{U} \in V$ matters: enlarging reconstructions beyond the consumer's reachable outputs could only lower the output-side function but is operationally meaningless, since every emitted $\hat{x}$ is read as $\mathcal{C}(\hat{x}) \in V$.

C. High resolution: the surrogate's exact validity region

Apply Thm. 4 first, so the residual problem lives on the output with local metric $G(u)$ and D_Y is a locally quadratic non-difference distortion in the sense of Linder–Zamir [25]: $q_G(u,\hat{u}) := \frac{1}{2}(\hat{u}-u)^\top G(u)(\hat{u}-u) = \|W(u)(\hat{u}-u)\|^2$, $W = (G/2)^{1/2}$, $D_Y = q_G + O(\|\hat{u}-u\|^3)$.

Theorem 5 (High-resolution closed form; finite-D pinch)**.** *Work in an intrinsic chart in which $U = \mathcal{C}(X)$ has an r-dimensional Lebesgue density with finite $h(U)$, taking h, G, and* det *in that chart, and assume the Linder–Zamir admissibility conditions [25] for (U, D_Y) (labels follow [25], whose (c),(d) do not bind here):* (a) *$D_Y(u,\cdot) \in C^3$ with uniformly bounded third-order partials;* (b) *a unique minimum at $\hat{u} = u$ with a piecewise-C^1, piecewise-invertible reproduction map;* (e) *$M(u) := \frac{1}{2}\nabla^2_{\hat{u}} D_Y(u,\hat{u})|_{\hat{u}=u} = G(u)/2 \succ 0$ a.e.;* (f) *the r-dimensional density just named;* (g) *$\mathbb{E}[\operatorname{tr}(G(U)^{-1})] < \infty$; and take V relatively open, containing a neighborhood of* $\operatorname{supp}(U)$*. Then:*

(i) *As $D \to 0$,*

$$R^{\text{true}}_{\mathcal{C}}(D) = h(\mathcal{C}(X)) - \frac{r}{2}\log\frac{2\pi e D}{r} + \frac{1}{2}\mathbb{E}\big[\log\det(G(\mathcal{C}(X))/2)\big] + o(1),$$

the input-weighted Linder–Zamir asymptotic ($R(D) = h(X) - \frac{n}{2}\log\frac{2\pi e D}{n} + \mathbb{E}\log|\det W(X)|$), which we verified to machine precision against exact reverse water-filling in the fixed-G Gaussian case. For KL-type consumers G is the Fisher metric and the middle term is a Jeffreys-prior log-volume: rate = entropy of what the consumer sees + log-volume of the consumer's metric + resolution*. The chart is essential: the ambient output metric may be degenerate—softmax's $G = \operatorname{diag}(p) - pp^\top$ has rank $r = S - 1$, so $\det G = 0$ in $\mathbb{R}^S$ and the formula is meaningful only in the $(S-1)$-dimensional simplex chart, where it further needs $\operatorname{supp}(U)$ bounded away from the boundary for (a),(g) to hold. The formula is chart-invariant as the combination $h(U) + \frac{1}{2}\mathbb{E}\log\det G$; genuinely* curved *$U$ (a nonlinear submanifold) needs a manifold extension the 1999 theory does not supply (Appendix B, §B-D).*

(ii) *If $\alpha\, q_G \le D_Y \le \beta\, q_G$ on $\operatorname{supp}(U) \times V$—a* global *two-sided sandwich, which for a merely local surrogate generally holds only near the diagonal / on bounded support, and so pairs with the high-resolution regime where $(U,\hat{U})$ concentrate—then for every $D > 0$, $R_{q_G}(D/\alpha) \le R^{\text{true}}_{\mathcal{C}}(D) \le R_{q_G}(D/\beta)$, by feasible-set inclusion; the bracket is computable at any D (Blahut–Arimoto; water-filling in the Gaussian case).*

Remark (What coding to the surrogate certifies). Thms. 4–5 replace Cor. 6's caveat (iii): coding to the point-dependent quadratic certifies the true downstream distortion *exactly* as $D \to 0$ and *within the* $[\alpha, \beta]$ *pinch* at any fixed D—the surrogate's validity region is quantified, not asserted. Since $d_{\mathcal{C}}$ is an admissible general distortion with a reference letter, the finite-blocklength dispersion theory of Kostina–Verdú [26] should port under their regularity conditions.

D. What remains open, and a registered prediction

Genuinely open after Thms. 3–5: *(a)* the curved-image case (U on a submanifold without an ambient density—the high-resolution theory needs charts); *(b)* the consumer-relative *channel* coding theorem behind §VI; *(c)* the multi-observer problem beyond the Gaussian weighted-quadratic case that §X now settles completely (region, refinability, rate loss, and the k-observer chain)—namely non-Gaussian sources, and the multi-observer region under the *true* nonlinear divergence, where the consumer quotients do not reduce to a common Gaussian program; and *(d)* non-i.i.d. sources.

GO-6 (proposed; to be sealed before running). At matched rate R, three coders: (a) *output coding*—compress $\mathcal{C}(X)$ under D_Y (Thm. 4-optimal); (b) *surrogate coding*—X-space, $d_{P_{\mathcal{C}}}$; (c) *reconstruction coding*—X-space MSE. Predictions: $d_{\mathcal{O}}(\mathrm{a}) \le d_{\mathcal{O}}(\mathrm{b}) \le d_{\mathcal{O}}(\mathrm{c})$, the (a)–(c) gap governed by the entropy share of $\ker P_{\mathcal{C}}$ and strict whenever that share is positive, the (a)–(b) gap vanishing as R grows (Thm. 5). The synthetic harness of §IV suffices.

Appendix C
The Price of a Misidentified Observer

The blind probe of §V recovers the read *subspace*, but imperfectly—on the trained Llama layer, at median overlap 0.647. This appendix prices that error, and finds a sharp asymmetry: mis-*weighting* directions the coder does read costs a bounded rate tax; *omitting* a direction the true operator reads erects a distortion floor no rate can cross. Throughout $X \sim \mathcal{N}(0, \Sigma_x)$, $\Sigma_x \succ 0$; $\Sigma^\star(P, D)$ is the max-det optimum of §X (Lemma 4); rates are in nats; $\tilde{P} := \Sigma_x^{1/2} P \Sigma_x^{1/2}$ is the whitened operator, and $R_P(D) = \frac{1}{2}\sum_{k:\, \tilde{p}_k > \theta} \log(\tilde{p}_k/\theta)$ with $\sum_k \min(\tilde{p}_k, \theta) = D$. The coder water-fills for an *estimated* $\hat{P}$ while the truth is P.

Theorem 6 (Commission tax)**.** *If $m\hat{P} \preceq P \preceq M\hat{P}$ on the shared range ($0 < m \le M$, which forces $\operatorname{range} P = \operatorname{range} \hat{P}$; let r be that rank), then for a true target D in the coding regime the design $\Sigma_\delta = \Sigma^\star(\hat{P}, D/M)$ guarantees $\operatorname{tr}(P\Sigma_\delta) \le D$, and its excess rate over the P-oracle obeys $0 \le R_{\hat{P}}(D/M) - R_P(D) \le \frac{r}{2}\log(M/m)$. The constant $r/2$ is sharp—attained when $P = m\hat{P}$ with all r modes active and only the loose M known.*

Proof. Safety: $P \preceq M\hat{P} \Rightarrow \operatorname{tr}(P\Sigma_\delta) \le M \operatorname{tr}(\hat{P}\Sigma_\delta) \le D$. Rate is Loewner-monotone in the operator and $R_{cP}(D) = R_P(D/c)$; so $P \preceq M\hat{P}$ gives $R_P(D) \le R_{\hat{P}}(D/M)$ ($\Rightarrow$ gap ≥ 0) and $P \succeq m\hat{P}$ gives $R_P(D) \ge R_{\hat{P}}(D/m)$. The gap is thus $\le R_{\hat{P}}(D/M) - R_{\hat{P}}(D/m)$; the sensitivity $dR/dD = -1/(2\theta)$ with $\theta \ge D/r$ (at most r active modes, each $\le \theta$) integrates to $\int_{D/M}^{D/m} \frac{r}{2D'}\, dD' = \frac{r}{2}\log(M/m)$. $\square$

Theorem 7 (Omission floor)**.** *If $\operatorname{range} P \not\subseteq \operatorname{range} \hat{P}$, let Π project onto the* whitened *kernel $\Sigma_x^{-1/2} \ker \hat{P} = \ker(\Sigma_x^{1/2}\hat{P}\Sigma_x^{1/2})$ and $D_{\text{floor}} := \operatorname{tr}(\tilde{P}\Pi)$. Then every allocation in the $\hat{P}$-water-filling family, at every rate, has true distortion in the compact band $D_{\text{floor}} \le \operatorname{tr}(P\Sigma_\delta) \le \operatorname{tr}(P\Sigma_x)$, with $D_{\text{floor}} > 0$ iff $\operatorname{range} P \not\subseteq \operatorname{range} \hat{P}$. Distortion is never unbounded, but reaching any $D < D_{\text{floor}}$ costs infinite rate—the rate diverges as $D \downarrow D_{\text{floor}}$.*

Proof. In the $\hat{P}$-whitened eigenbasis $\Sigma_\delta(\theta) = \Sigma_x^{1/2} S(\theta) \Sigma_x^{1/2}$, $S(\theta) = 1$ on $\ker(\Sigma_x^{1/2}\hat{P}\Sigma_x^{1/2})$ and $\min(1, \theta/\hat{p}_k) \in [0, 1]$ on read modes, so $\Pi \preceq S(\theta) \preceq I$; tracing against $P \succeq 0$ gives the band. As $\theta \to 0$, $S(\theta) \to \Pi$ and $\operatorname{tr}(P\Sigma_\delta) \to D_{\text{floor}}$ while the read-block rate diverges. Positivity: $\operatorname{tr}(\tilde{P}\Pi) > 0$ iff $\ker(\Sigma_x^{1/2}\hat{P}\Sigma_x^{1/2}) \not\subseteq \ker \tilde{P}$, i.e. (whitening is a bijection) $\operatorname{range} P \not\subseteq \operatorname{range} \hat{P}$. The whitened kernel is the correct one: in whitened coordinates the modes are independent, so the residual on the unread block is its full variance Π; the naive $\ker \hat{P}$ overstates the floor when $\Sigma_x \ne I$. $\square$

From a measured overlap to a certified floor. The probe of §V reports the subspace overlap $o = \frac{1}{r}\operatorname{tr}(\Pi_R \Pi_{\hat{R}})$ between the true read subspace $R = \operatorname{range} P$ and the recovered $\hat{R}$. Under the projector model $P = \Pi_R$, $\hat{P} = \Pi_{\hat{R}}$, $\Sigma_x = I$, Theorem 7 gives $D_{\text{floor}} = r(1 - o)$: a fraction $1 - o$ of the read energy is irreducible. The measured median $o = 0.647$ (Llama-3.2-3B, §V) thus *certifies* a floor of 0.353 of the read energy—*provided the recovery omits the true subspace and the read weights are equal.* This is a conditional certificate, not a flat figure: Prop. 4 recovers the subspace but not the eigenvalues, so a weighted floor $\sum_i \lambda_i \sin^2\phi_i$ needs data the overlap does not carry, and an *over*-covering probe ($\hat{R} \supseteq R$) has $D_{\text{floor}} = 0$—it reads everything the true operator does. What gates the floor is the true read energy lying *outside* $\hat{R}$, which the symmetric overlap o does not isolate.

The keys catastrophe, modeled. The boundary result of §XI (GO-5) refutes the $\alpha = 1$ density quotient for a direct inner-product consumer: the density is *read* (not in $\ker P_{\mathcal{C}}$, Prop. 6), so, modeled within the quadratic-Gaussian water-filling picture, quotienting it acts as an $\hat{P}$ whose range omits a direction the true P reads—exactly Theorem 7's hypothesis, hence a positive floor and irrecoverable failure at any budget. The diffusion-domain case, where the density lies *in* $\ker P_{\mathcal{C}}$, is pure commission:

bounded wasted bits (Theorem 6), the observed $< 2\%$ effect. One knob, two regimes, separated exactly by whether the discarded direction is inside or outside $\operatorname{range} P$—the theory-side account the practice paper lacked.

We verified Theorems 6–7 numerically (sealed prereg GO-P-2026-024, fresh-context pass logged): the commission bound is tight at $P = m\hat{P}$, the floor equals $\operatorname{tr}(\tilde{P}\Pi)$ with rate diverging as $D \downarrow D_{\text{floor}}$, and the naive kernel overstates the floor by $\sim$30% when $\Sigma_x \neq I$. A sealed, held-out prospective test (GO-P-2026-027, layers $\{4, 20\}$) confirms the floor operationally on all 16 KV heads: the read-metric floor is reached, and the infinite-rate misidentified coder plateaus at a softmax divergence 800–5700$\times$ that of the matched operator (median 0.10 nats)—a genuine downstream wall, out-of-sample from the layers on which the effect was first seen. A finite-rate sweep initially missed it (the reducible read-mode error dominates the swept rates), so the floor is a high-resolution phenomenon whose finite-rate visibility scales with the omitted direction's downstream weight; the consumer's own metric inherits it in the high-rate limit.

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All results resolve to committed registry entries and reproduction artifacts in the public companion repository [22]; each registry entry is committed *before* the run it governs, so registration precedes experimentation in the commit ordering of the public history. That ordering is not cryptographic third-party proof; an external immutable deposit (Zenodo) accompanies the paper.

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